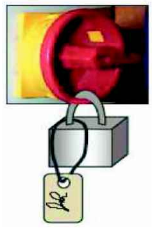
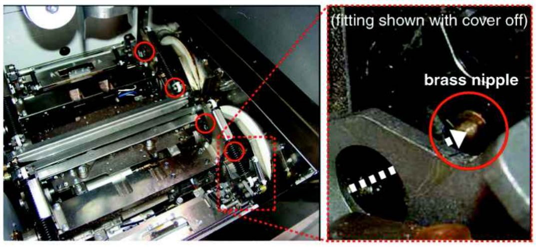
WARNING

- **Jaws of the Bagmaker may be very hot. Allow the jaws to cool before working in the jaw area.**
  Use caution when working near the knife blade.

1. Follow all Lockout / Tagout Procedures.



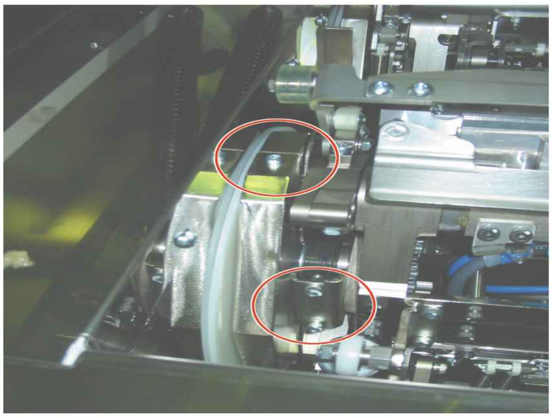
2. Open the front cover.
3. Locate the four brass cam follower lubrication nipples.



CAUTION

- **Use caution rotating jaws manually to avoid pinch points.**

4. Remove the cam follower covers (clam shell type) by removing two 3mm Allen bolts and the four 5mm Allen bolts.

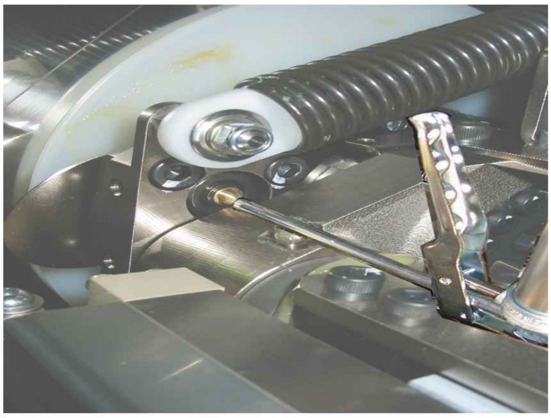


NOTE

- Rotate the jaws 180° to locate the opposite set of bolts.
  The 5mm bolts are firmly secured. Use an extension handle to loosen them.

5. After removing the Allen bolts, use your hand to remove both clam shell covers.
   Completely clean the front and real jaw clam shell covers.

6. Once the covers have been cleaned and slid into position, replace the Allen bolts.
7. Place the Ishida grease gun into position and slide the nozzle over brass nipple.
8. Pump the grease gun 5 times. **Rotate jaws 180° to lube second front nipple.**

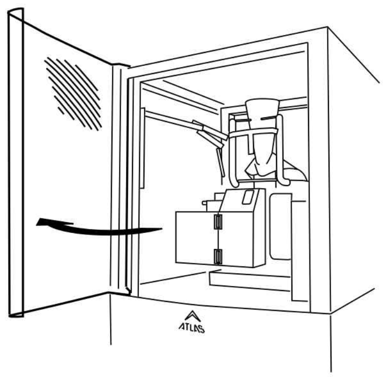


9. Repeat the procedure on the rear jaw cam followers top and bottom nipples.

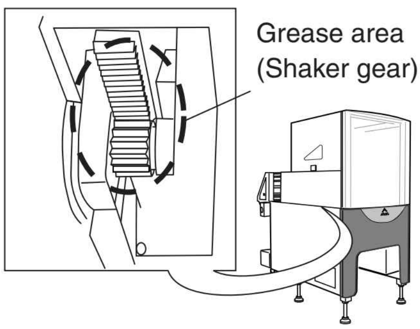
3.2.1.4 Shaker Unit Gear (Option)

1. Press the [RUN/LACE] key.

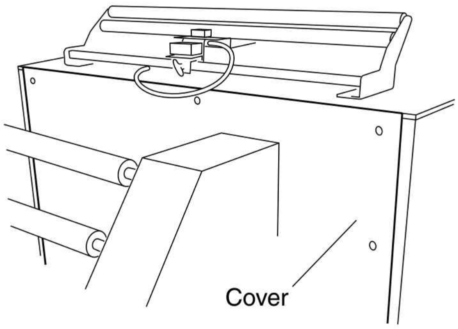
   ▶The pull-down belts move outward.
2. Turn the main power OFF.
3. Open the front cover.



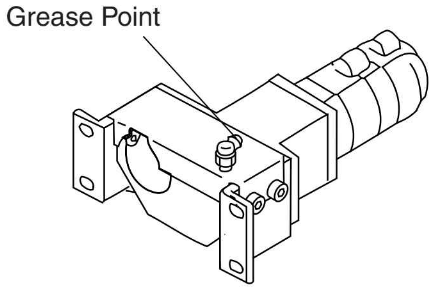
4. Grease the inner gear.



5. Open the rear upper cover by removing ten screws.



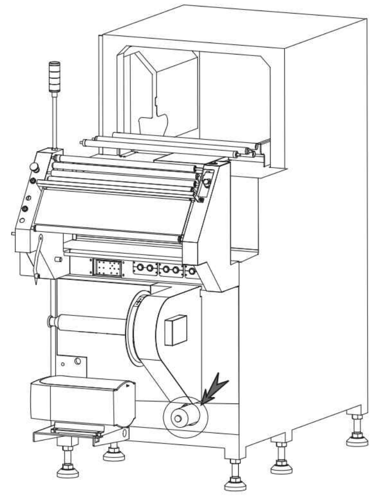
6. Grease the outer gears.



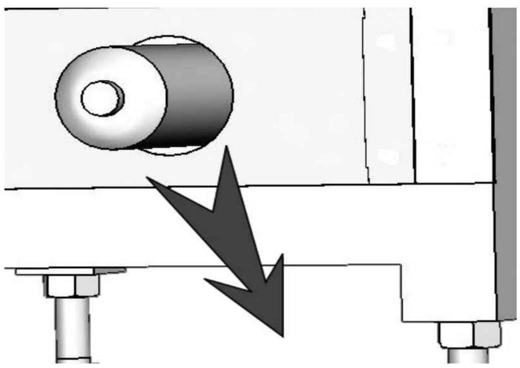
3.3 6-Month Inspections

3.3.1 Vacuum System Air Filter

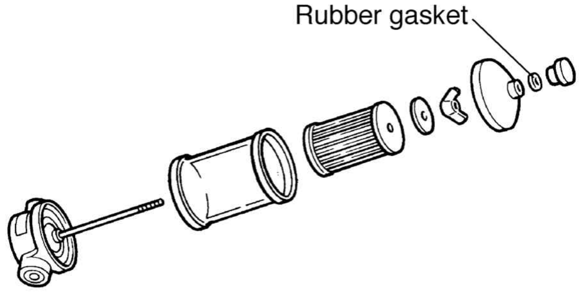
1. Turn the main power OFF.



2. Remove the air filter.
3. Using an air gun, blow out anything strange that may be adhering to the air filter.



4. Attach the air filter.
5. Attach the outer cover.



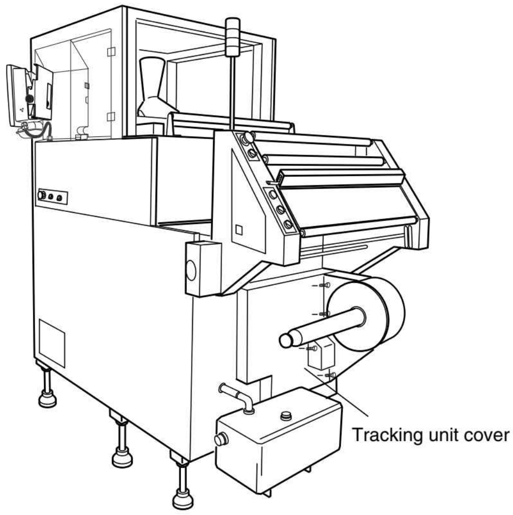
3.3.2 Lubrication

CAUTION

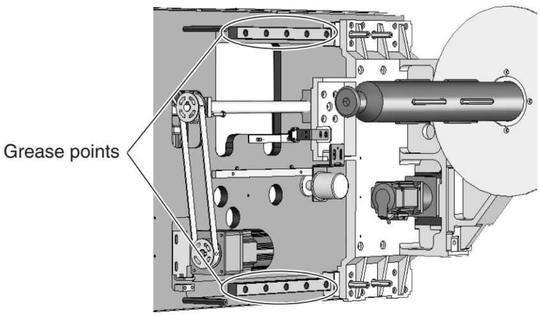
- **Lubricate periodically.**

3.3.2.1 Film Tracking Linear Guides

1. Turn the main power OFF.
2. Remove the film tracking unit cover.



3. Grease the film tracking unit linear guides.



4 REPLACEMENT AND ADJUSTMENT

4 Replacement and Adjustment

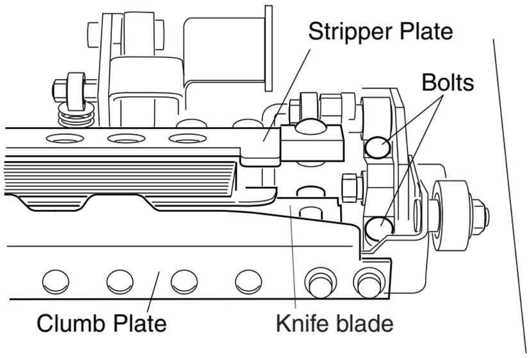
DANGER

- **Before opening the control panel cover, turn the main power OFF. Failure to do so can result in electrical shock.**

4.1 End Seal Unit

4.1.1 Replacing the Jaw Faces

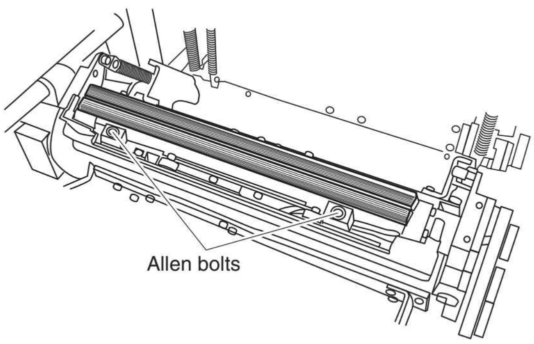
1. Turn the main power OFF.
2. Open the front cover.
3. Pull the handle forward, and swing the back seal unit out.
4. Remove 4 bolts and remove the crumb plate and the stripper plate.
5. Remove the allen bolts and remove the knife blade.



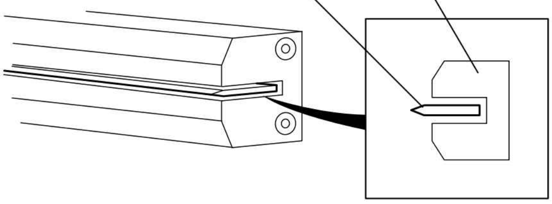
WARNING

- **Be careful when removing the knife blade. It is very sharp and hot.**

6. Replace the front and rear jaw faces with new ones.



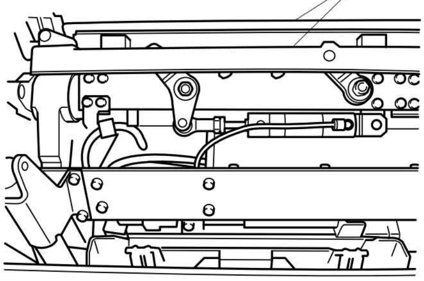
- Install the front jaw face so that the knife blade is in the middle of its knife slot.
- Install the rear jaw face temporary.



7. Make sure that the front and rear jaw faces are properly aligned.

   If the alignment is improper, adjust the position of the jaw. (For details on how to adjust the jaw, see "4.1.2 Jaw Alignment")

8. Tighten the allen bolts of the rear jaw.



4.1.2 Jaw Alignment

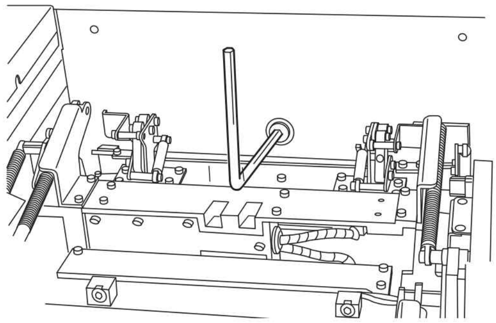
WARNING

- **It is sometimes necessary to have the main power ON when making adjustments. At such times, when the main power is ON, make sure that the motor power supply is turned OFF. If the motor power supply is left ON, the motor could start up, injuring someone. In addition, wear protective gloves when adjusting electrical equipment.**

1. Jog the jaw until the front and rear jaws engage.
2. If the following cases appear, jaw adjustment will be needed.
   a. Jaw alignment has upward or downward deviation.
   b. Jaw alignment has play.
   c. Jaw alignment has horizontal deviation.

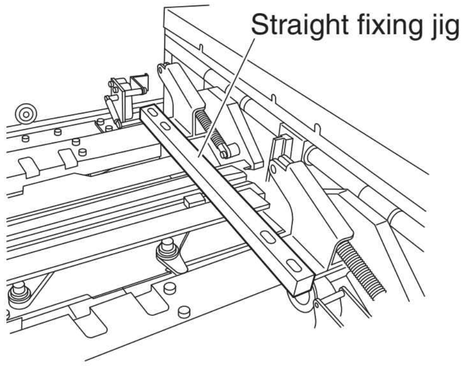
   To adjust the Jaw alignment, follow the below steps.

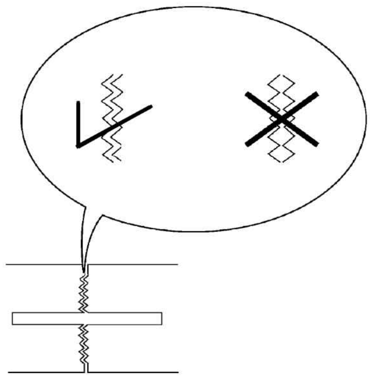
1. Turn the main power OFF.
2. Open the front cover.
3. Pull the handle forward, and swing the back seal unit out.
4. Remove 4 bolts and remove the crumb plate and the stripper plate.
5. Turn the pivot point screw to close jaw.



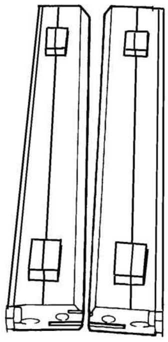
6. Adjust the front and rear jaw to match the knife.
7. Attach a straight fixing jig to keep jaw horizontal.
8. Check alignment with mirror.

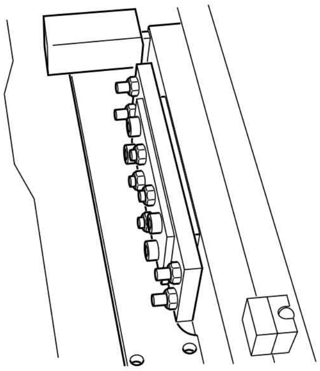
   If the jaw alignment is loose, adjust as follows.



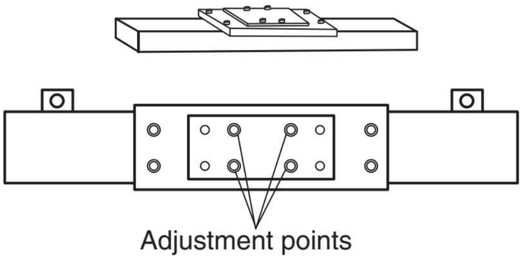


Right or Left adjustment

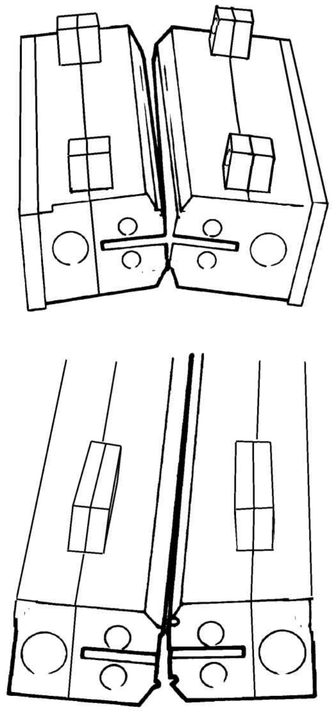




If the right and left are not aligned, adjust by turning adjustment screws on the metal plate attached to rear jaw.

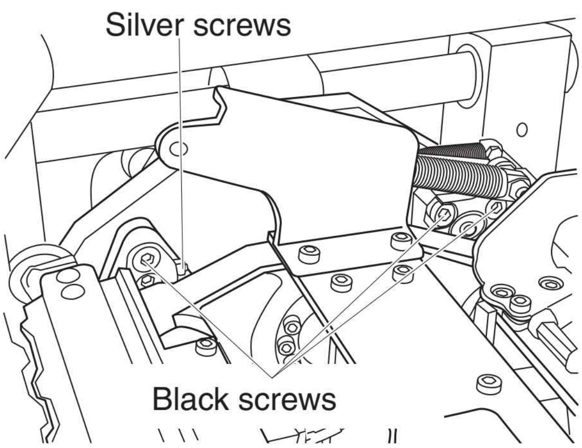


Top and Bottom adjustment



If top and bottom are not aligned, adjust the screw on the arm.

1. Loosen the 3 black screws.
2. Tighten or loosen the silver screw to adjust jaw face.



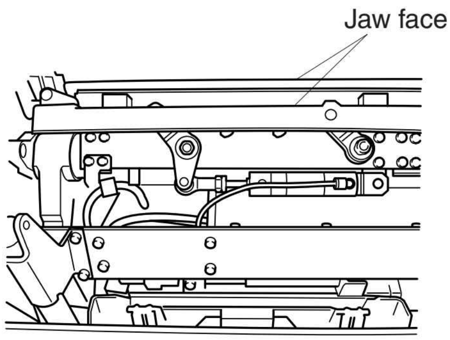
NOTE

- There are two sets of jaw, so check both of jaw alignment.

4.1.3 M2 Motor

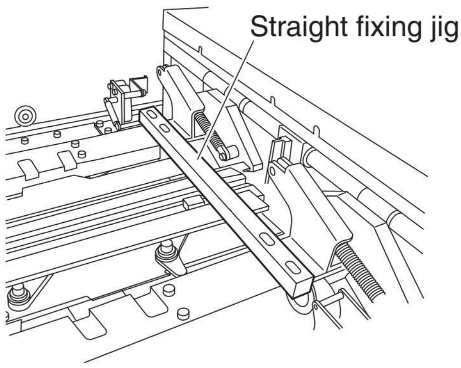
4.1.3.1 Timing Belt of M2 Motor

1. Proceed steps 1 to 4 of the section "4.1.1 Replacing the Jaw Faces".

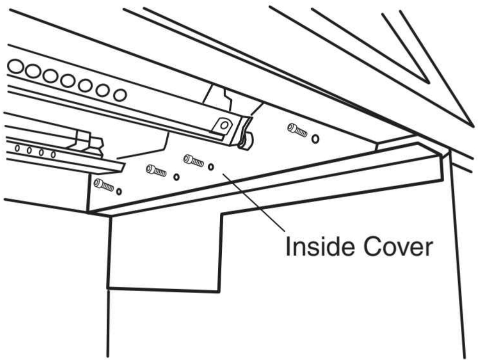


2. Using your hand, rotate the jaw arm and engage the front and rear jaws.

3. Attach a straight fixing jig to the jaw arm.



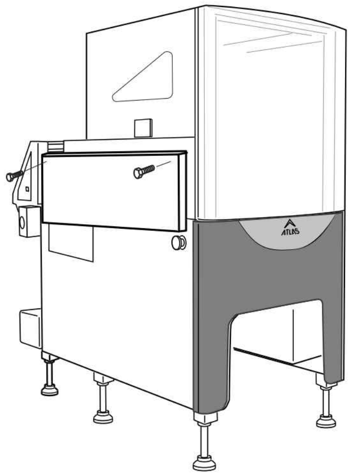
4. Remove 4 screws, remove the inside cover.



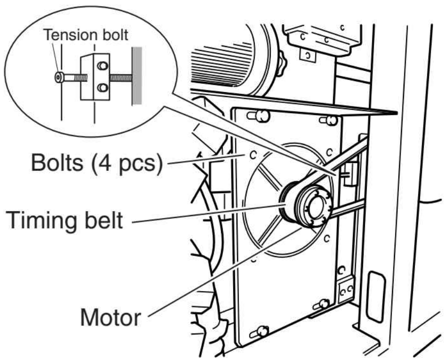
5. Remove the LEFT side cover.

CAUTION

- **Two people should remove the cover, since it is heavy.**

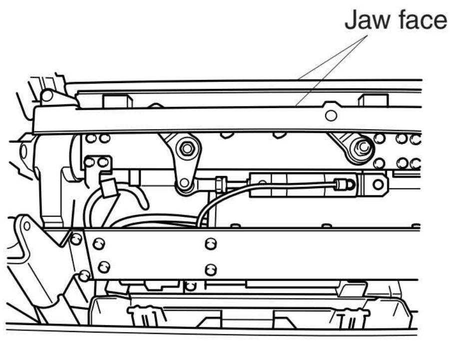


6. Loosen 4 bolts tightening the motor bracket.
7. Loosen the tension bolt.
8. Remove the timing belt.
9. Install the new timing belt.
10. Using your finger, press down on the center of the belt between pulleys. Adjust until the deflection is approximately 2 millimeters then tighten the tension bolt and tighten the motor bracket screws.
11. Attach the inside cover.

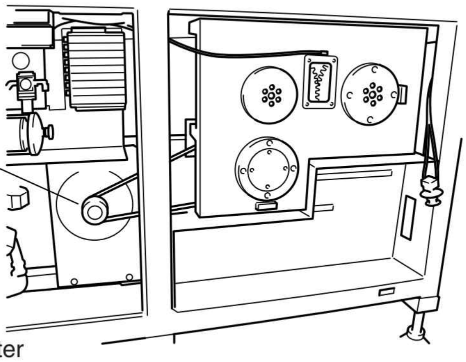


4.1.3.2 Replacing the M2 Motor

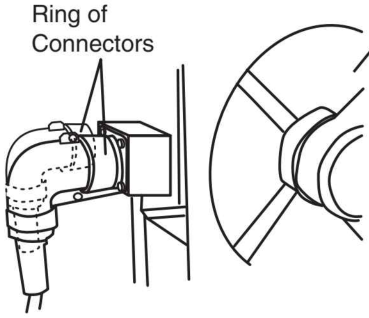
1. Proceed steps 1 to 4 of the section "4.1.1 Replacing the Jaw Faces".
2. Using your hand, rotate the jaw arm and engage the front and rear jaws.
3. Attach a straight fixing jig to the jaw arm.



4. Remove 4 screws and remove the noise filter unit so that you can access the M2 motor.



5. Loosen the ring, remove two connectors at the right of M2 motor.
6. Loosen 4 bolts tightened the motor bracket.
7. Loosen the tension bolt.
8. Remove 4 screws, remove the inside cover.
9. Remove the timing belt.
10. Loosen the mechanical lock, and remove the pulley from the M2 motor.
11. Remove the allen bolts, and remove the M2 motor.



CAUTION

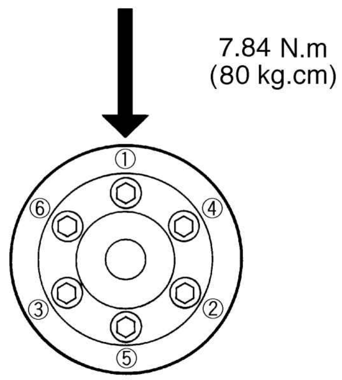
- **The motor is heavy and should be lifted by two people.**

12. Remove the motor from the motor base.
13. Attach the motor to the new motor base.

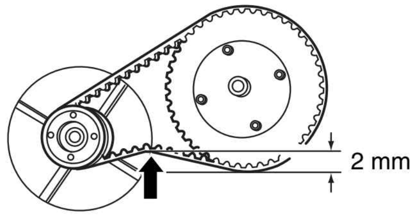
CAUTION

- **Do not cause any shock to the motor, or the motor's encoder can be damaged.**

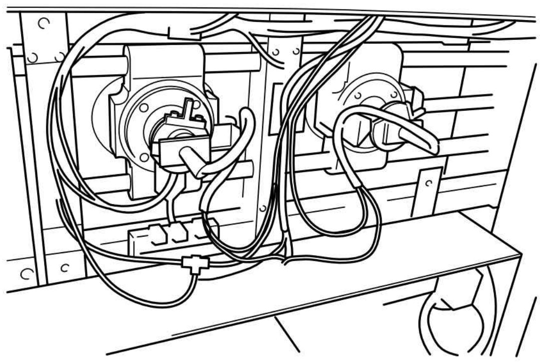
14. Attach and fasten the gear and mechanical lock to the motor shaft. Tighten the bolts for the mechanical lock at 7.84 N.m (80 kgf.cm) in a crisscross (star) pattern.
15. Install the motor with motor base.
16. Install the electrical box of the PS-0 unit.



17. Attach the timing belt.
18. Using your finger, press down on the center of the belt between gears. Adjust until the deflection is approximately 2 millimeters; then fasten the M2 motor.
19. Tighten the tension bolt.
20. Attach the inside cover.



4.1.3.3 Home Position of Jaw Rotation

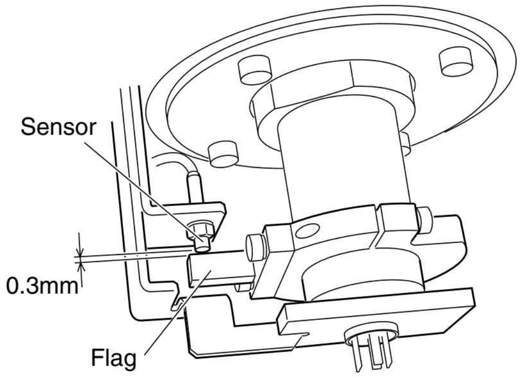


Home Position is the position where the jaw stops after the motor is turned on.
Then, either knife 1 or knife 2 comes upper side.

- When the rear sensor turned on, knife 1 comes to upper side.
- When the front sensor turned on, knife 2 comes to upper side.
- When Home position, the plate of the jaw should be angled 90 degree against the horizontal frame of main unit.

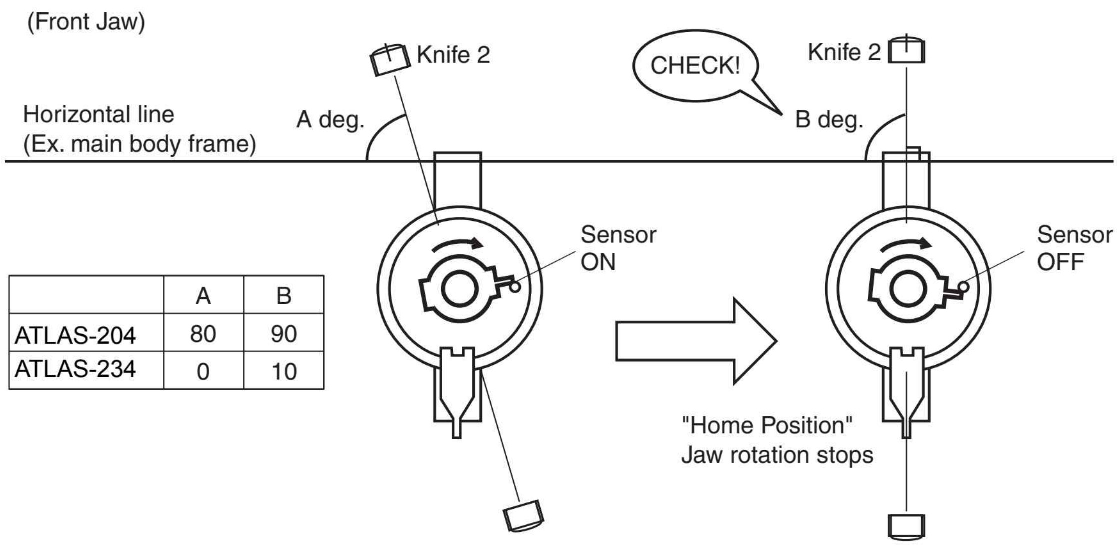
1. Turn the motor power supply OFF.
2. Turn the main power supply OFF.

3. Open the left lower cover.
4. Set gap to 0.3mm between sensor and flag.



5. Set the angle of the jaw plate to A degree against the horizontal frame of main unit when the sensor turns on.
6. Turns the motor power on.

   Jaw rotates until the sensor switches on, then the jaw stops.
7. Confirm if the angle of the jaw plate is B degree against horizontal frame.



	A	B
ATLAS-204	80	90
ATLAS-234	0	10

8. If the angle is not B degree, adjust the flag, then restart the motor power on, and confirm the angle when the jaw stops at home position.

NOTE

- Repeat the above step until the angle to be correct.

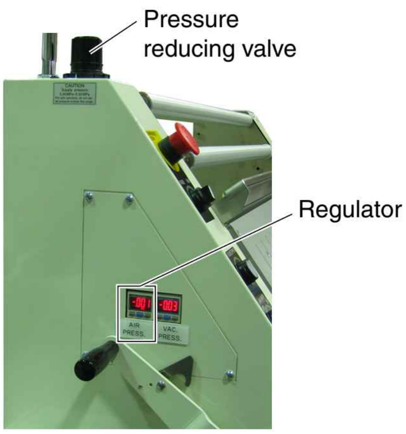
9. Adjust both of the Knife 1 and Knife 2 jaw sensor.

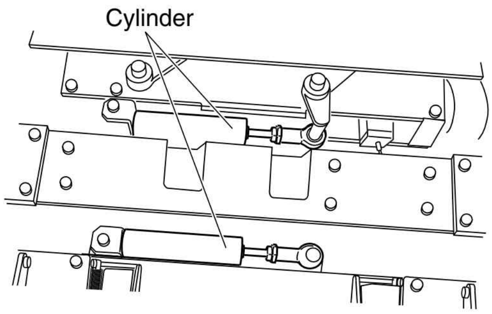
<How to adjust the flag>

1. Loosen the screws tightening the flag, and rotate the flag to appropriate angle.
2. Tighten the screws.

4.1.4 Replacing the Air Cylinder for Knife

1. Turn the main power OFF.
2. Open the front cover.
3. Pull the handle forward, and swing the back seal unit out.
4. Remove 4 bolts and remove the crumb plate and the stripper plate.
5. Pull the knob for the pressure reducing valve and adjust until the pressure is 0.0 MPa.
6. Remove the rod end ; then replace the cylinder for knife with a new one.

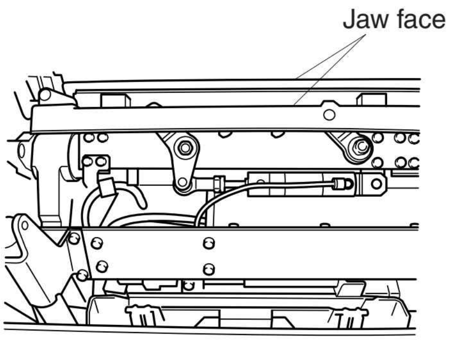




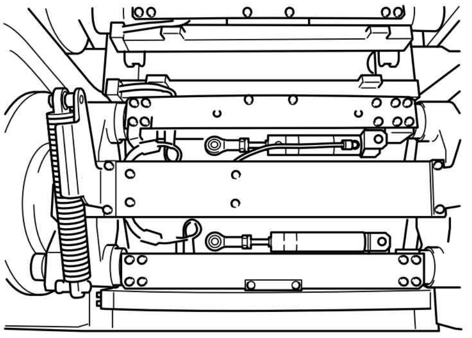
4.1.5 Jaw Heater

4.1.5.1 Replacing the Jaw Heater

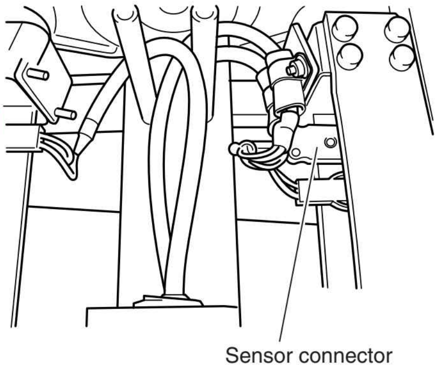
1. Turn the main power OFF.
2. Open the front cover.
3. Pull the handle forward, and swing the back seal unit out.
4. Remove 4 bolts and remove the crumb plate and the stripper plate.



5. Remove the sensor connector of the end seal heater from the plate.



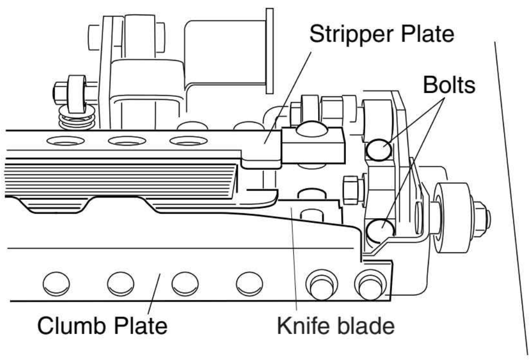
6. Disconnect the cable that runs from the terminal connector for the heater.



CAUTION

- **The thermocouple has polarity and is identified by the color of the cable (red or white).**
  If a connection is made with the wrong polarity, the temperature control can fail or unit can be damage.

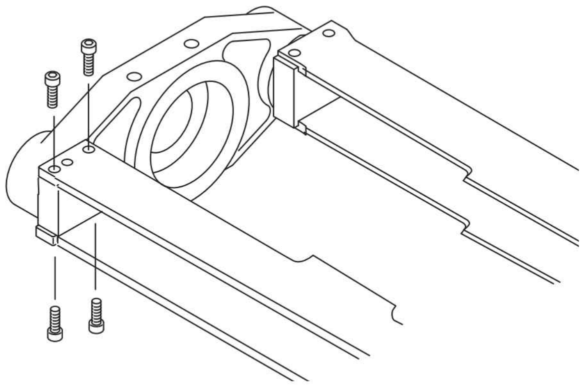
7. Remove the allen bolt and remove the knife blade.



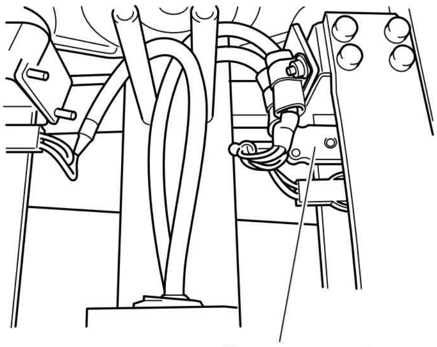
WARNING

- **Be careful when removing the knife blade. It is very sharp and hot.**

8. Remove 8 screws, remove the heater unit from the arm unit.



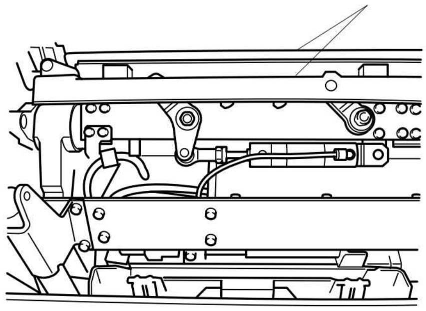
9. Remove 4 screws, remove the heater cover.
10. Remove 3 screws, remove the heater cap then pull the heater.



TIP

- In case of replacing thermocouple, remove the fixing screw then pull the thermocouple.

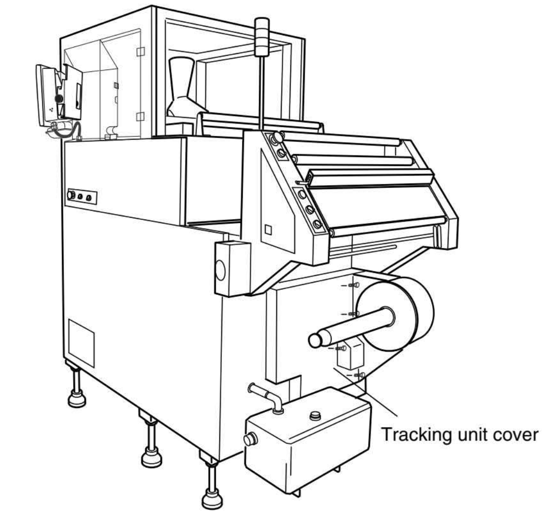
11. Attach the new heater or thermocouple.
12. Install in the reverse order of removal.
13. Check the engagement of the jaws. If the engagement is not proper, adjustment is needed. (To adjust the position of the jaws, see "4.1.2 Jaw Alignment")



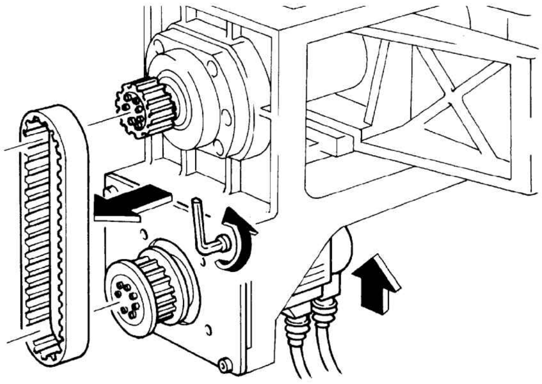
4.1.6 M3 Motor

4.1.6.1 Timing Belt of M3 Motor

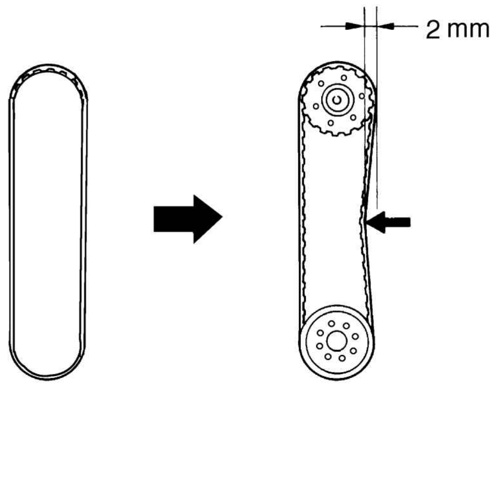
1. Turn the main power OFF.
2. Remove the tracking unit cover.
3. Move the tracking unit to the right edge.
   (Rotate the tracking motor by hand to move the unit.)
4. Perform the operation from the back side of the main unit.



5. Loosen the allen bolts that fasten the M3 motor.
6. Lift the M3 motor, and temporarily fasten the allen bolts.
7. Replace the belt with a new one.
8. Lift the motor down and tighten the bolt.

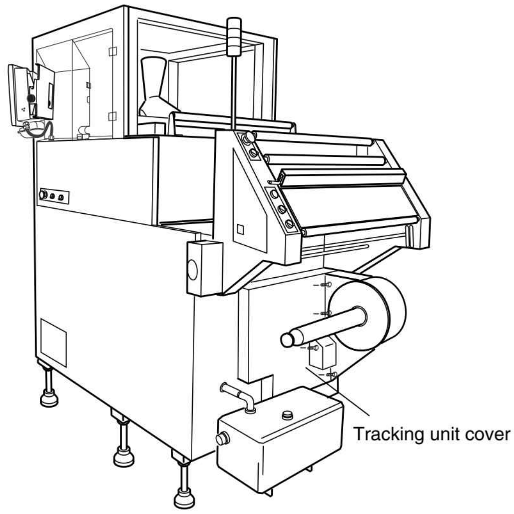


9. Using your finger, press down on the center of the belt between pulleys. Adjust until the deflection is approximately 2 millimeters; then tighten the M3 motor fixing bolts.

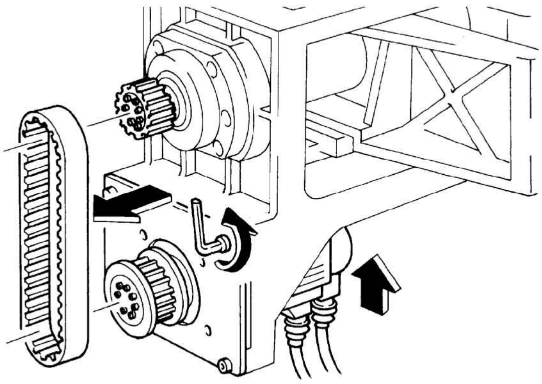


4.1.6.2 Replacing the M3 Pivot Point Motor

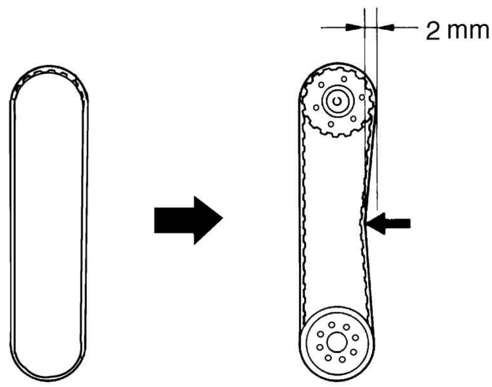
1. Turn the main power OFF.
2. Remove the tracking unit cover.
3. Remove the main unit lower cover on the right side.
4. Move the tracking unit to the right edge. (Rotate the tracking motor by hand to move the unit.)



5. Loosen the allen bolts that fasten the M3 motor.
6. Lift the M3 motor, and temporarily fasten the allen bolts.
7. Remove the timing belt.
8. Remove the two cables connected to motor.
9. Remove the 4 screws, and replace the motor with a new one.
10. Move the electrical control panel with the front side of the panel centered and remove the motor through the panel for installation.

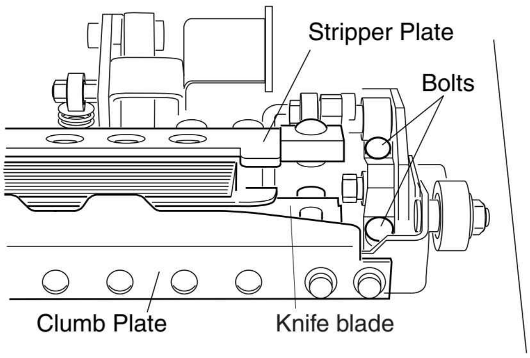


11. Temporarily fasten the motor.
12. Attach the belt.
13. Lift the motor down.
14. Using your finger, press down on the center of the belt between gears. Adjust until the deflection is approximately 2 millimeters; then tighten the M3 motor fixing bolts.



4.1.6.3 Home Position of Pivot

1. Turn the motor power supply OFF.
2. Turn the main power supply OFF.
3. Remove the main unit lower cover on the right side.
4. Open the front cover.
5. Pull the handle forward, and swing the back seal unit out.
6. Remove 4 bolts and remove the crumb plate and the stripper plate.



CAUTION

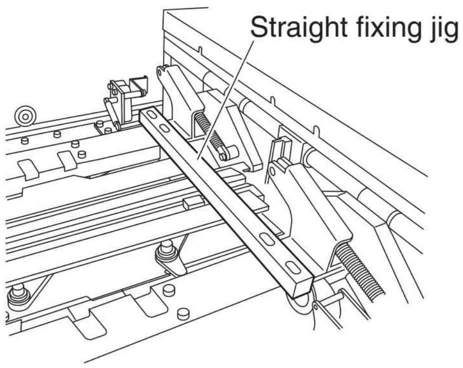
- **After operation, the jaws are extremely hot. Do not touch directly. You can get burned.**

7. Jog arms around until jaw faces mesh together and arm are perfectly horizontal.

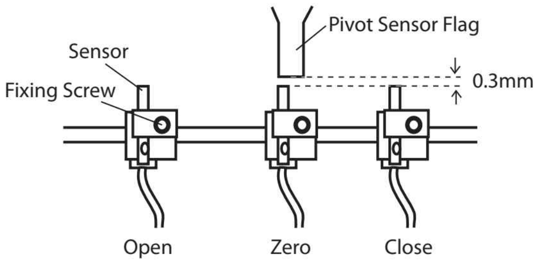
NOTE

- Arms must be perfectly horizontal.

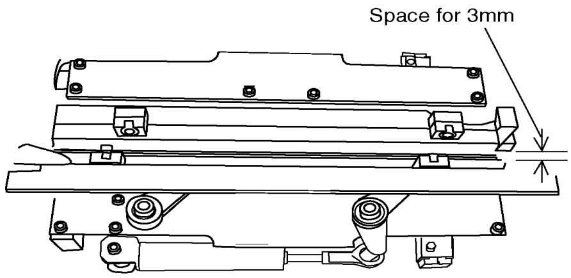
8. Attach a straight fixing jig to the jaw arm.



9. Set gap to 0.3mm between flag and sensor.



10. Make a space for 3mm between jaw faces.

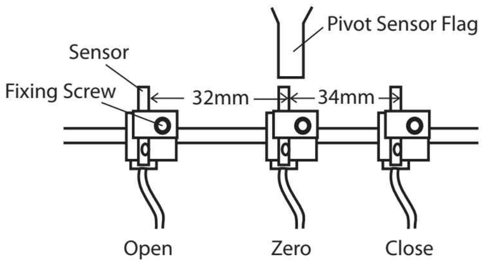


11. Adjust the middle(zero) sensor position to the right edge of the flag.

12. Remove the fixing jig.

13. Set the distance to 34mm between middle block and right(close) block.

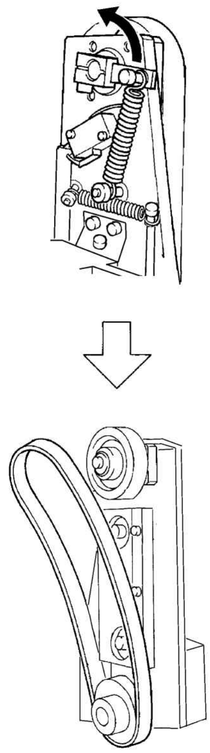
14. Set the distance to 32mm between middle block and left(open) block.



4.2 Back Seal Unit

4.2.1 Replacing the Heater Band

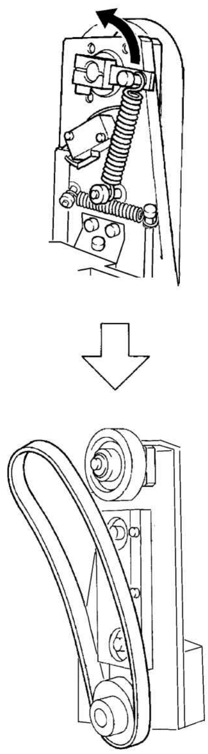
1. Turn the main power OFF.
2. Open the front cover.
3. Pull the handle forward, and swing the back seal unit out.
4. Open the back seal cover.
5. Lift the tension lever located on the side of the back seal unit.
6. Replace the heater band with a new one.



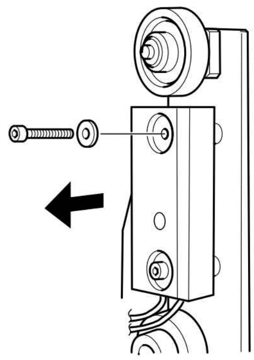
4.2.2 Back Seal Heater

4.2.2.1 Replacing the Heater Sensor

1. Turn the main power OFF.
2. Open the front cover.
3. Pull the handle forward, and swing the back seal unit out.
4. Open the back seal cover.
5. Lift the tension lever located on the back of the steel belt, and remove the heater belt.



6. Remove the heater block.

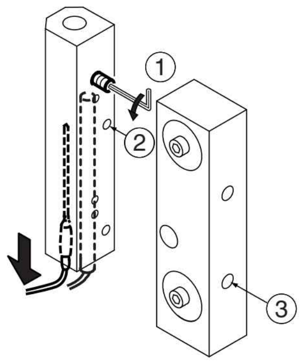


NOTE

- The lengths of the bolts that fasten the heater block are different.

7. Loosen the two allen screws ③ from the side of the heater block and remove the heater shoe.

8. Loosen the screws ① and/or ② from the side of the heater block, and pull out the heater and/or thermocouple.



CAUTION

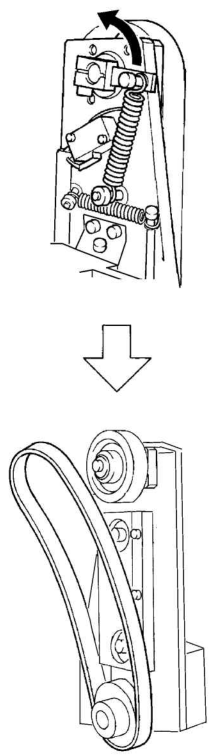
- **If the thermal joint is sticking, making it difficult to pull out the heater, press with a rod-shaped tool from the opposite side; then pull out.**

9. Replace the heater and/or thermocouple with new ones.

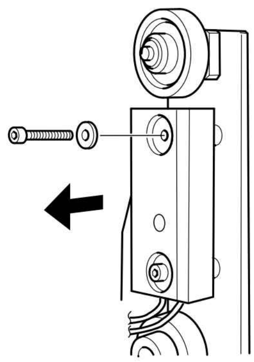
10. Reinstall all parts in the reverse order of removal.

4.2.2.2 Replacing the Heat Pipe

1. Turn the main power OFF.
2. Open the front cover.
3. Pull the handle forward, and swing the back seal unit out.
4. Open the back seal cover.
5. Lift the tension lever located on the back of the steel belt, and remove the heater belt.



6. Remove the heater block.

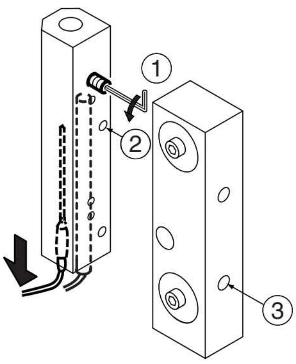


NOTE

- The lengths of the bolts that fasten the heater block are different.

7. Remove the two bolts that fasten the heater shoe to the heater block.

8. Loosen the two allen screws ③ from the side of the heater block and remove the heater shoe.



9. Loosen the screws ① and/or ② from the side of the heater block, and pull out the heater and/or thermocouple.

10. Install the new heat pipe.

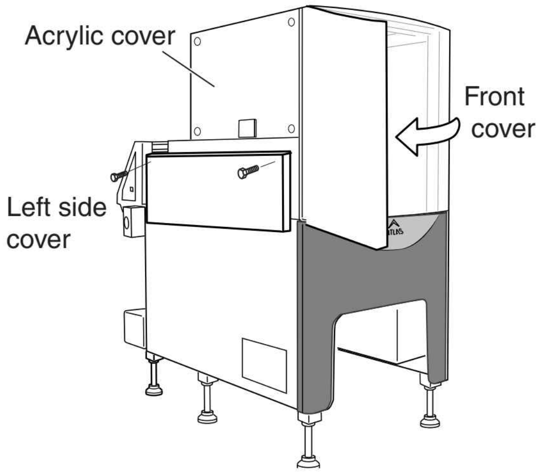
11. Reinstall all parts in the reverse order of removal.

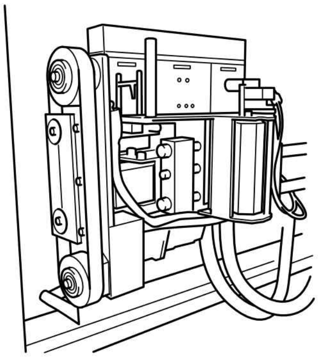
4.2.3 Replacing the Back Seal Air Cylinder

NOTE

- You must remove the back seal unit to access the back seal cylinder.

1. Turn the main power OFF.
2. Lift the pressure reducing valve knob; then turn the knob counterclockwise until the pressure is at 0.
3. Open the front cover, left upper cover and acrylic cover.
4. Pull the handle forward, and swing the back seal unit out.

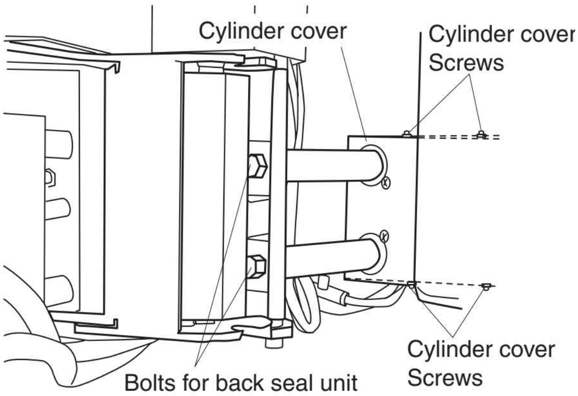




5. Remove the two bolts fixing the back seal unit, remove the back seal unit.

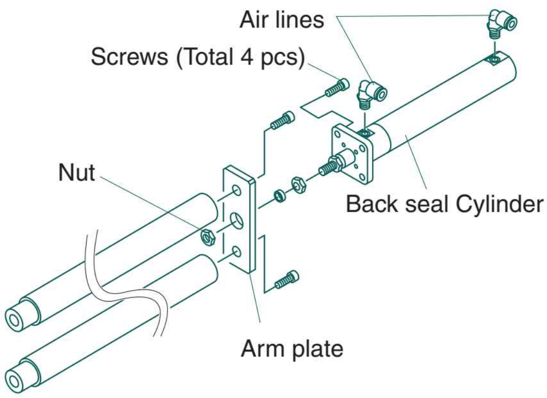
NOTE

- The back seal unit is heavy so put on the space carefully not causing any shock.
- The back seal unit removal is just to access to the back seal cylinder. Back seal unit is out of replacing items.



6. Remove each two screws on the top and bottom sides of cylinder cover and remove the cylinder cover.

7. Remove a nut fastened arm plate and remove the cylinder.
8. Remove the two air lines from the fitting.
9. Remove 4 screws and remove the cylinder from the main body.
10. Install the new air cylinder.
11. Reinstall all parts in the reverse order of removal.



4.2.4 Replacing the Back Seal Unit M5 Motor

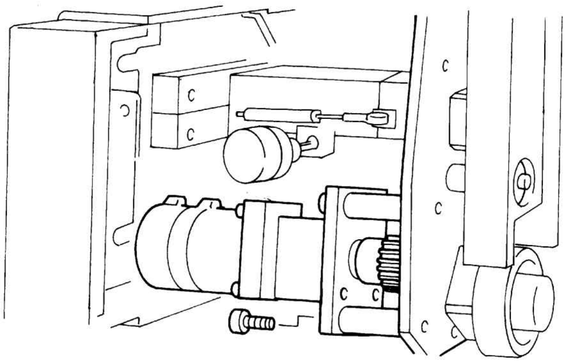
1. Turn the main power OFF.
2. Open the front cover.
3. Open the back seal unit.
4. Remove the arm cover.
5. Remove the CN sig cable from the M5 connector.

CAUTION

- **Pull the cable out straight while holding the connector. Failure to do so can damage the cable.**

6. Remove the M5 servo motor horizontally. The M5 motor is fastened by four allen bolts.

   Tool: 5-mm hex wrench

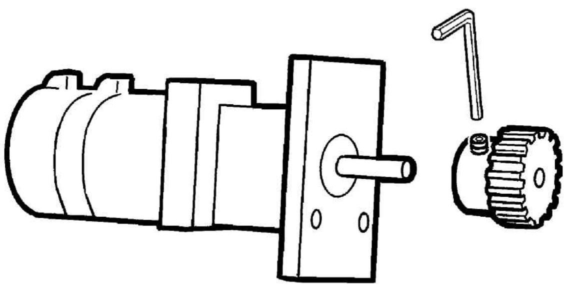


NOTE

- Removing the guard plate will make the replacement work easier.

7. Remove the gear from the M5 servo motor.

   Tool: 2.5-mm hex wrench



NOTE

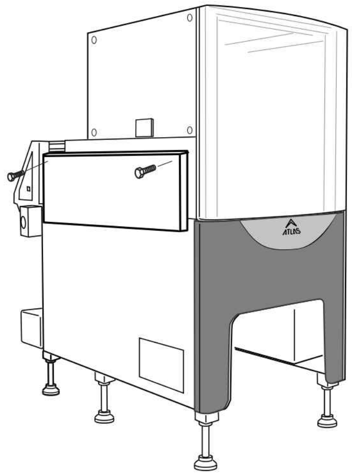
- Replace the motor only. The gears are not included.

8. Attach the original gear to the new M5 servo motor.
9. Attach the M5 servo motor to the back seal unit.
10. Reinstall all parts in the reverse order of removal.

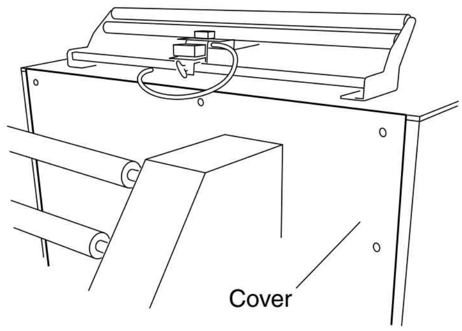
4.3 Pull Belt Unit

4.3.1 Replacing M1 Motor of the Pull Belt

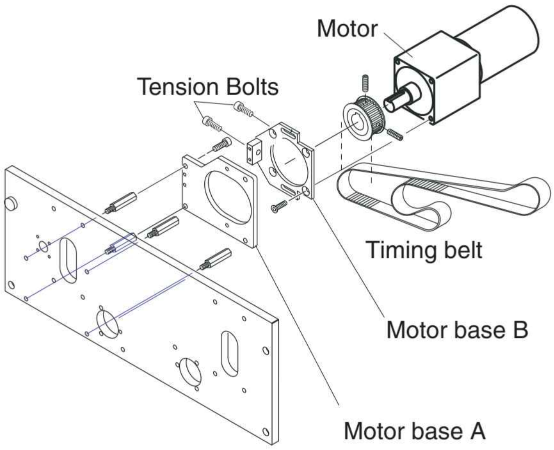
1. Turn the main power OFF.
2. Open the left side upper cover.
3. Remove the 2 back side screws fixing the left side upper cover, then remove the front side 2 screws to remove the cover.



4. Remove the ten screws, remove the rear upper cover.



5. Loosen two tension bolts.
6. Remove 2 screws and remove the motor base A from the M1 motor with motor base B.
7. Remove 4 screws, remove the M1 motor from the motor base B.
8. Remove the timing belt if necessary.



NOTE

- You cannot remove timing belt without removing the motor.

9. Reinstall all parts in the reverse order of removal.
10. Check for proper tension on pull belts.

    Deflection : 3-4mm

4.3.2 Replacing the Pull Belt

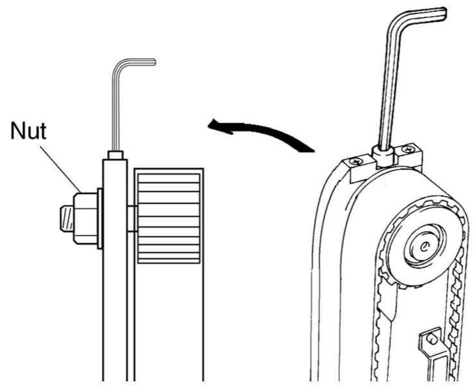
1. Press the [Run/Lace] key.

   ▶The pull-down belts move outward.
2. Turn the main power OFF.
3. Open the front cover.
4. Pull the handle forward, and swing the back seal unit out.
5. Remove the former, and make sure that the pull belt is free of grime, flaws, and splits.

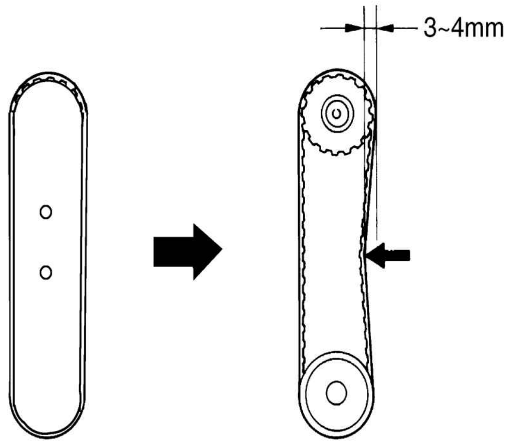
CAUTION

- **Be careful when handling the former. It is heavy.**

6. Loosen the nut at the reverse side of the pull belt unit and loosen the tension.
7. Remove the pull belt.
8. Install the new pull belt.



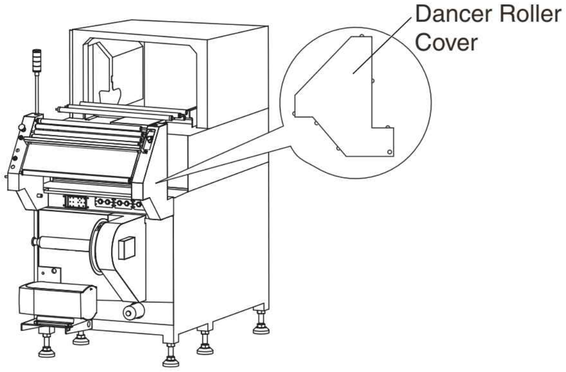
9. Press down on the center of the pull belt and adjust the tension that the belt deflects approximately 3 to 4 millimeters, then tighten the nut.



4.4 Dancer Roller

4.4.1 Replacing the Load Cell

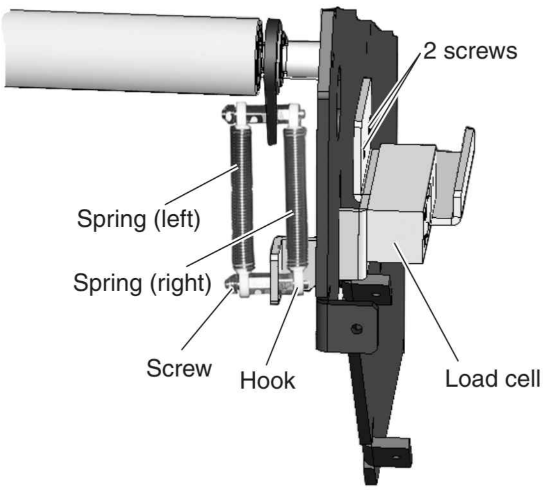
1. Turn the main power OFF.
2. Remove the main unit lower cover on the right side.
3. Move the electrical control panel to remove the load cell connector in the back.



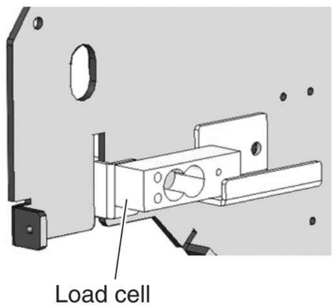
4. Remove the Dancer Roller cover on the left side.
5. Remove the dancer roller spring.

   Left: Remove the nut attaching the bottom of the spring.

   Right: Remove the 4 screws of the load cell move the load cell and detach the spring from the hook.
6. Remove the 2 bolts fixing the load cell fitting outside the frame.

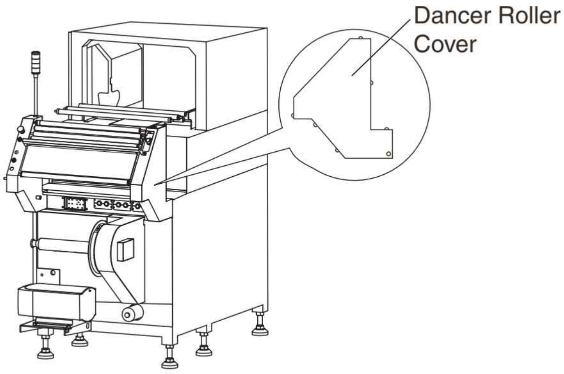


7. Remove the load cell.



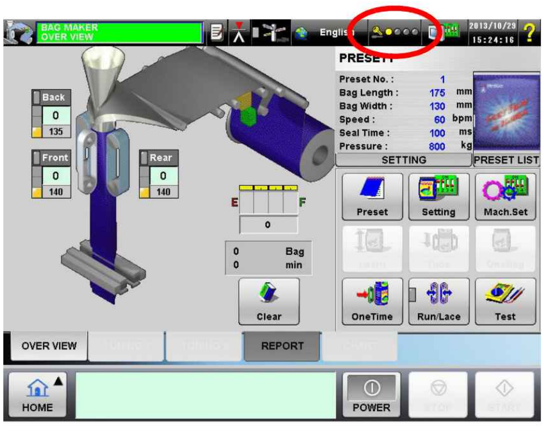
4.4.2 Dancer roller adjustment

1. Turn the main power supply ON.
2. Remove the Dancer Roller cover on the left side.



3. Press the operation level key  on a screen.

   ▶The screen to select the operation level will appear.

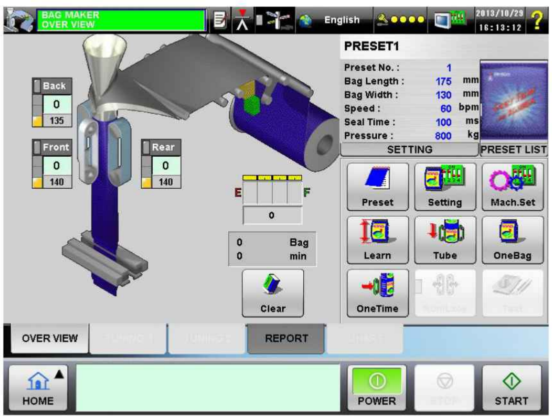


4. Select the Maintenance level.

   ▶The keyboard to enter the password will appear.

5. Enter the password for the Maintenance level.

   ▶The Operation Standby screen for the Maintenance level will appear.



6. Press the 

 key.

   ▶The MACHINE SETTING screen will appear.

7. Press the [SYSTEM SETUP] index.

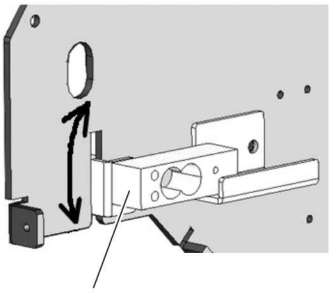
   ▶The keyboard to enter the password will appear.

8. Enter the password "3720".

   ▶The SYSTEM SETUP screen will appear.

9. Confirm that the following values have been displayed for the load cell data on the screen:

   If dancer roller is at lower limit: 520 (±10)
   If dancer roller is at upper limit: 4000-4400



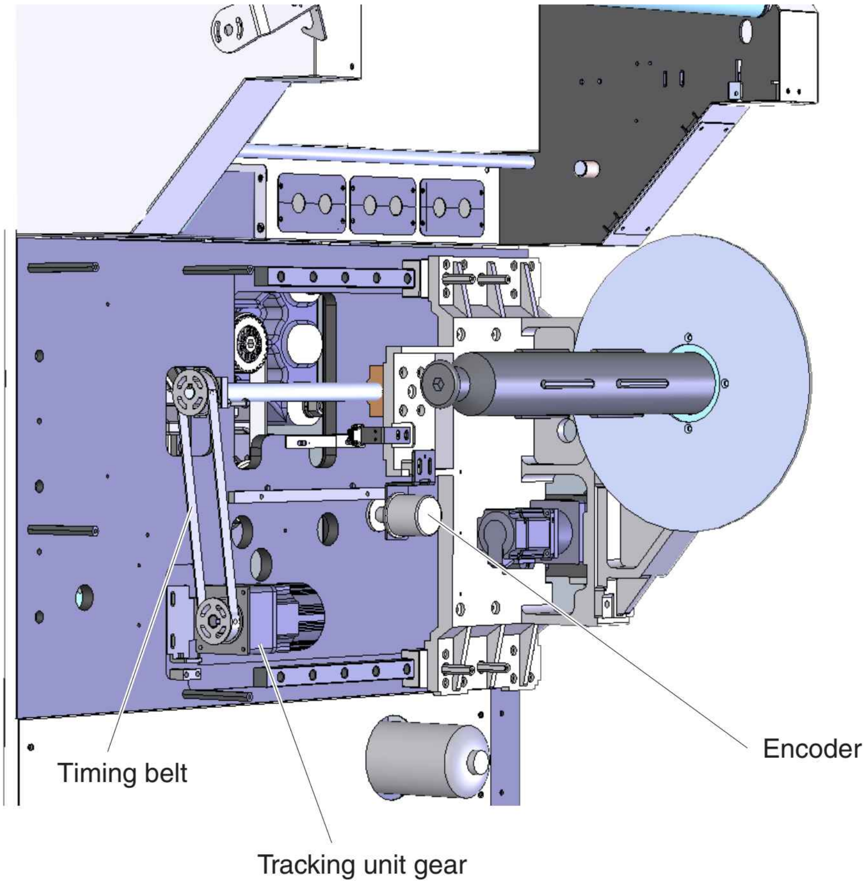
NOTE

- If the specified values are not given, change the angle of the load cell for adjustment.

10. Loosen the fixing screws, then loosen the bolts from the inside to adjust the position.

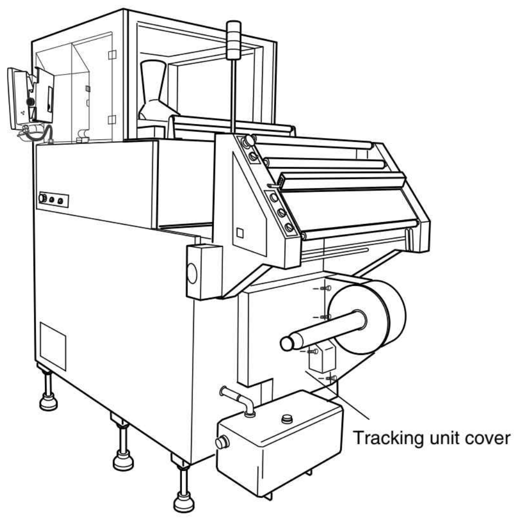
11. Attach the dancer roller cover.

4.5 Tracking Unit

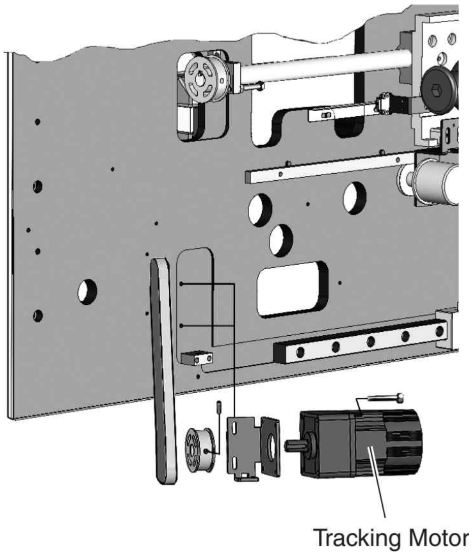


4.5.1 M7 Tracking Motor

1. Turn the main power OFF.
2. Remove the back side cover.
3. Loosen 4 bolts of the motor bracket.
4. Loosen the tension of the tracking motor.

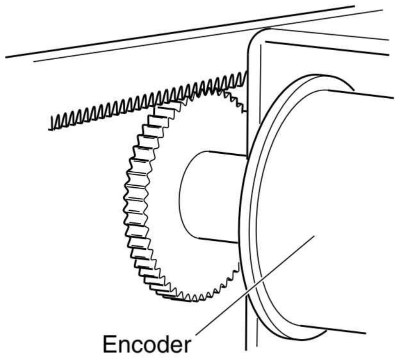


5. Remove the timing belt.
6. Remove the motor bracket by removing 4 bolts.
7. Remove the head gear from the motor.
8. Remove 4 bolts and remove M7 motor from the bracket.
9. Attach the head gear to the new motor.
10. Install the motor with attached head gear.
11. Reinstall all parts in the reverse order of removal.



4.5.2 Replacing the Encoder

1. Turn the main power OFF.
2. Remove the back side cover.
3. Remove the two screws and remove the encoder with bracket.
4. Remove the encoder gear.
5. Remove the 4 screws and remove the encoder from the bracket.



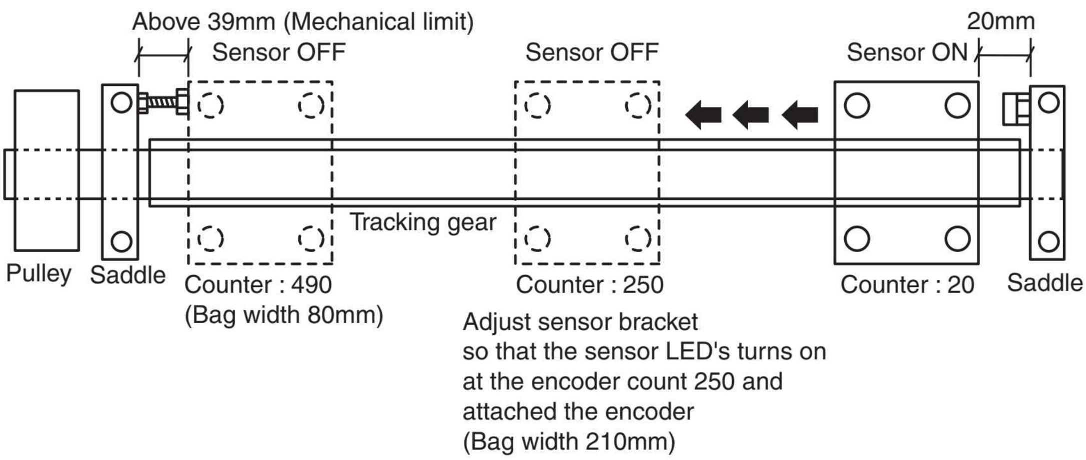
4.5.3 Tracking unit Adjustment

NOTE

- If you need replacement, replace the parts according to the parts list.

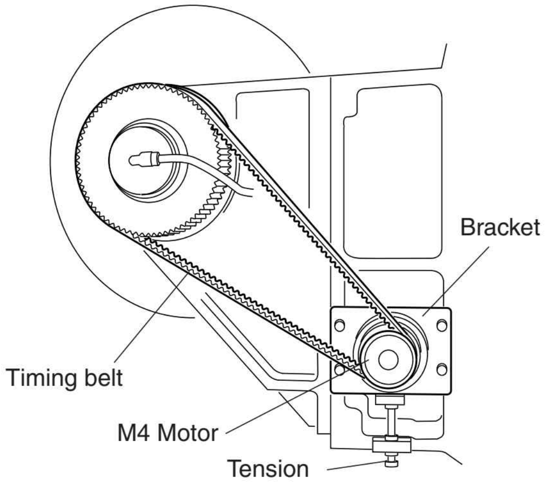
Tracking unit Adjustment

1. Turn the main power supply ON.
2. Remove the Tracking unit cover.
3. Switch to the remote control screen's tracking adjustment.
4. Set the encoder to count "20" at 20mm from the saddle without pulley.
5. Move the tracking unit with rotating the pulley till the encoder counts 250.
6. Adjust L plate so that the tracking sensor turns off at this position.
7. Additionally, check if the encoder counts 490 at the position of above 39mm from the saddle which is the next of pulley).



4.6 Replacing the Film Unwind Unit

4.6.1 Replacing the M4 Motor



1. Turn the main power OFF.
2. Remove the film spindle unit cover.
3. Loosen 4 bolts of the motor bracket, and loosen the tension bolt.
4. Detach the timing belt.
5. Remove 4 screws and remove the motor with bracket.
6. Remove the belt gear.
7. Remove the M4 motor from the bracket.
8. Attach the original gear to the new motor; then apply a small amount of graphite grease.

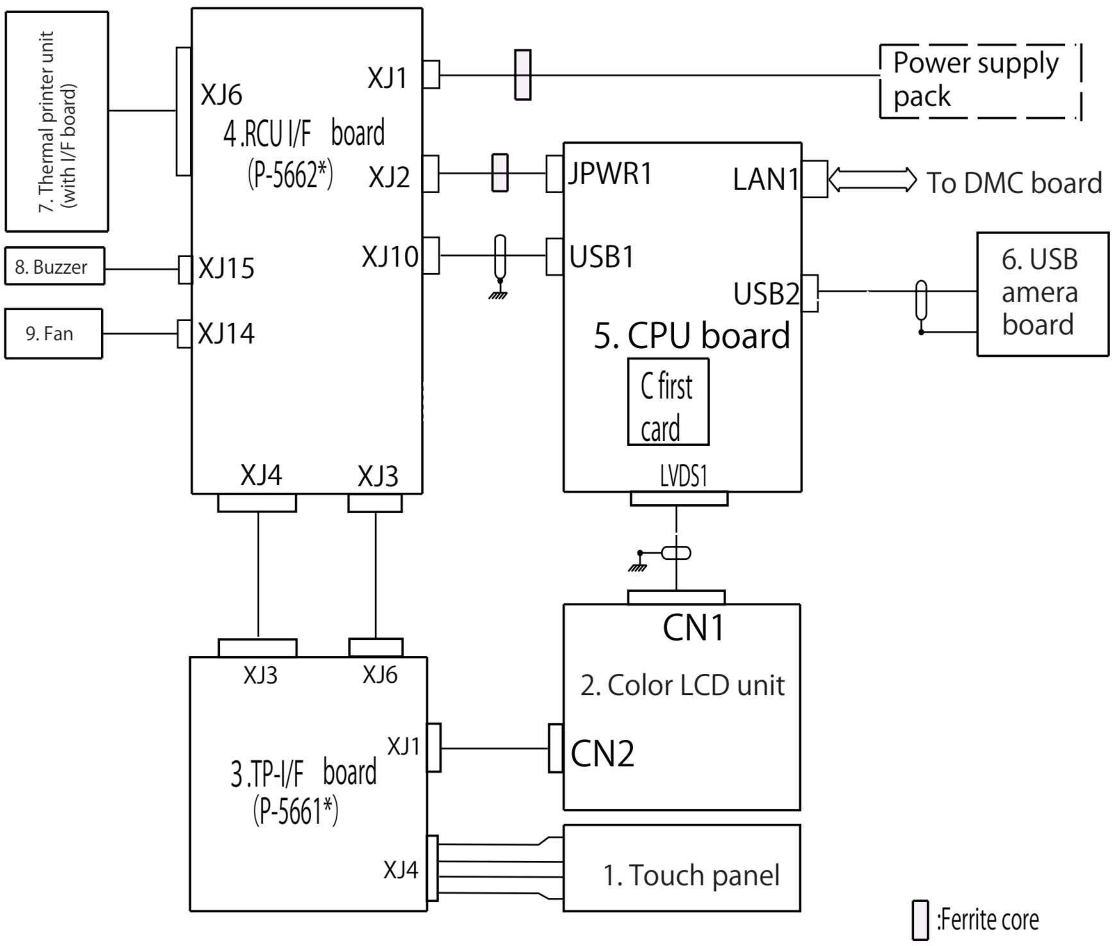
NOTE

- Replace the motor only. The gears are not included.

9. Install the new motor.
10. Reinstall all parts in the reverse order of removal.

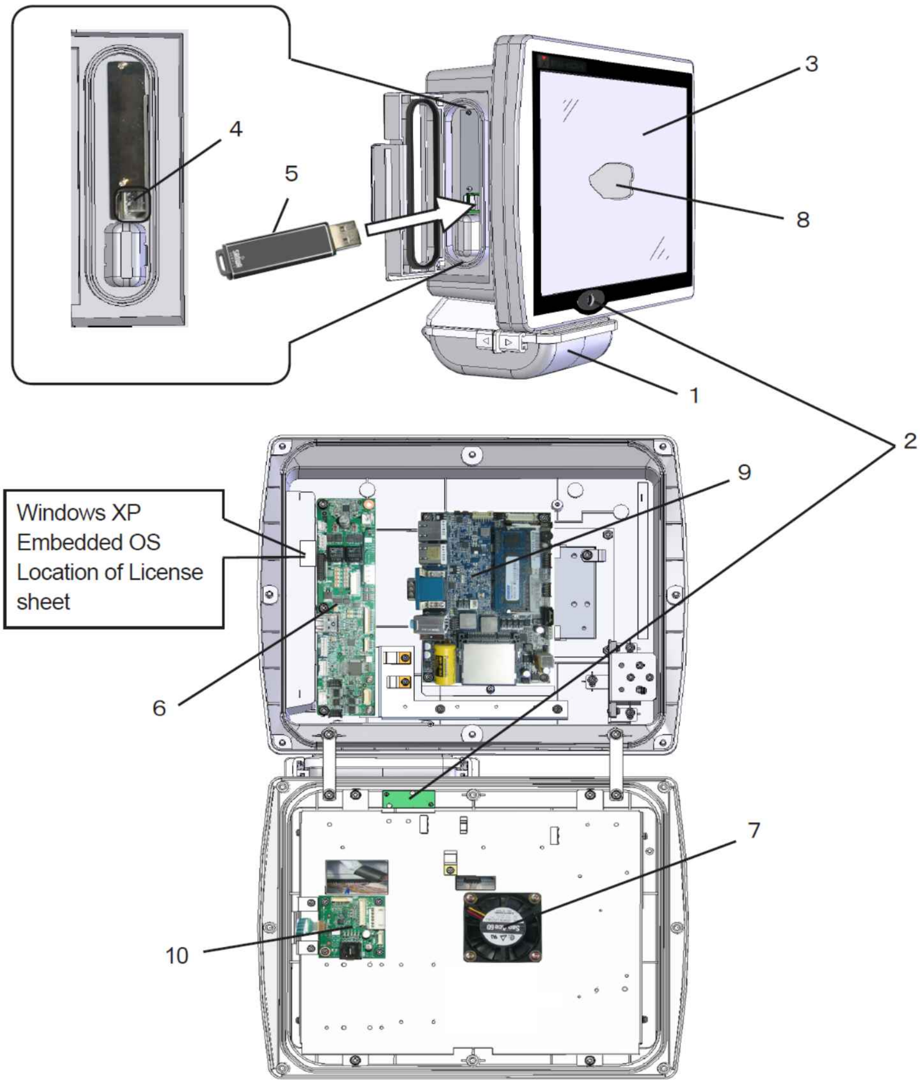
4.7 RCU Unit (option)

4.7.1 Remote Control Unit Block Diagram



Symbol	Description	Type no. / Notes	Maker
1	Touch panel	KBT-12.1C10R-FM	KB Elctron
2	Color LCD unit	AC121SA01	Mitsubishi
3	TP-I/F board (P-5661*)		
4	RCU-I/F board (P-5662*)		
5	CPU board	NPC-M0103	Omron
6	USB camera board		
7	Printer	SAM-1245-10K	SEIKO
8	Buzzer		
9	Fan		
10	USB camera switching board (P-5582*)	(option)	

4.7.2 Remote Control Unit Outline View



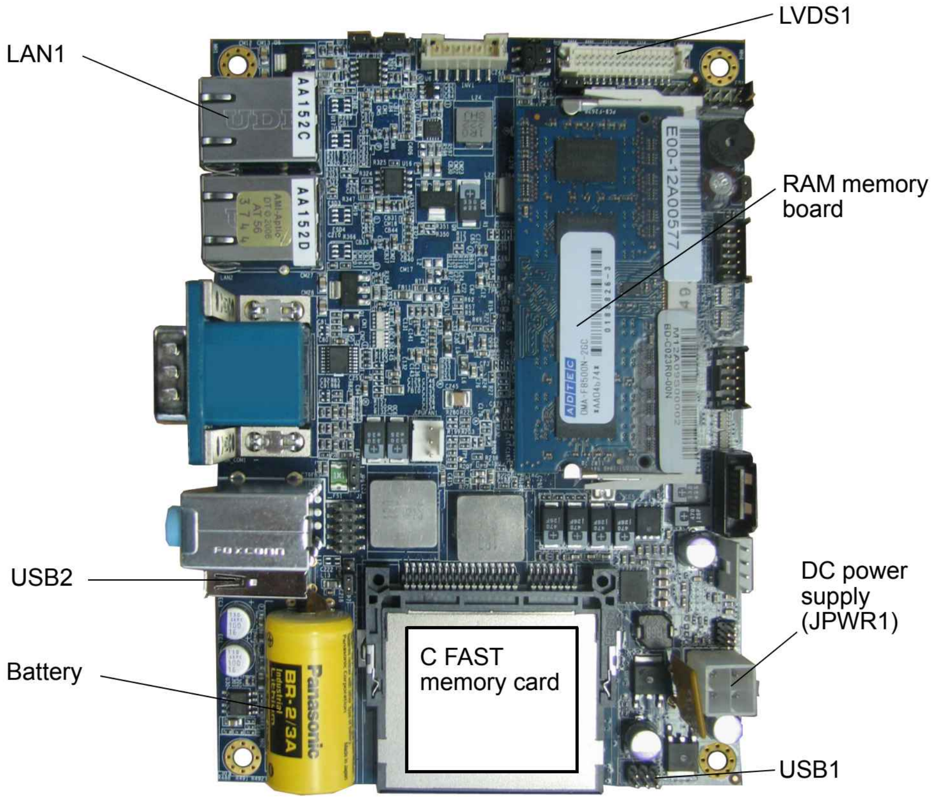
No.	Unit Description
1	Printer unit
2	USB camera
3	Touch panel
4	USB memory slot (2 slots)
5	USB memory
6	RCU I/F board (P-5662*)
7	Fan motor
8	Color LCD unit
9	CPU board (NPC-M0103) (OMRON)
10	TP-I/F board (P-5661*)

4.7.3 CPU Board (NPC-M0103)

WARNING

- **Before replacing electrical components, make sure that the main power supply and motor power supply are turned OFF. Failure to do so can result in electrical shock.**

Ourline and Connector location



Function of board

1. Input control by touch panel
2. Communication [Ethernet] with CAL [DMU board (P-5562 *)]
3. Communication [USB1 slot] with RCU I/F board (P-5562*)
4. Controls USB camera, USB camera selection board (P-5562*)

Battery

1. Battery in use: Lithium primary battery (accumulator battery) Model: BR-2/3A
2. Replacement time: 15 years (It depends on the environment.)

CAUTION

- **The battery replacement with a wrong one may cause malfunction in the board. Replace with the same type or equivalent.**
- **Discard the used battery.**

Connector classification

Connector No.		Function
LAN1	(8P)	Ethernet communication (to DMU board)
LVDS1	(30P)	LCD control
JPWR1	(4P)	DC power supply (+12 V)
USB1	(6P)	USB communication (to RCU I/Fboard)
USB2	(4P)	Control of USB camera, USBcamera switching board

Connector function in details

(1) JPWR1(DC-IN): Power supply input

Connector No.	Pin No.	Signal name	Remarks
DC-IN (JPWR1)	1	GND	
	2	GND	
	3	+12 V	
	4	+12 V	

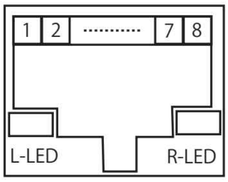
(2) USB1:USB (internal connection) interface

Connector No.	Pin No.	Signal Name	Pin No.	Signal Name
USB1 Interface (forInternal connection)	1	VCC_USB	2	GND
	3	USBON	4	GND
	5	USBOP	6	(N.C.)

(3) USB2:USB (externmal connection) interface

Connector No.	Pin No.	Signal name	Remarks
USB2 Interface (for External connection)	1	VCC_USB	
	2	USBN	
	3	USBP	
	4	GND	

(4) LAN1: LAN interface

Connector No.	Pin No.	Signal name	Remarks
LAN1 Interface	1	MDI_OP	
	2	MDI_ON	
	3	MDI_1P	
	4	MDI_2N	
	5	MDI_2P	
	6	MDI_1N	
	7	MDI_3P	
	8	MDI_3N	

L-LED

Green : ON	100 M
Orange : ON	Giga
OFF	10 M

R-LED

Yellow : blink	Act
Yellow : ON	Link
OFF	OFF

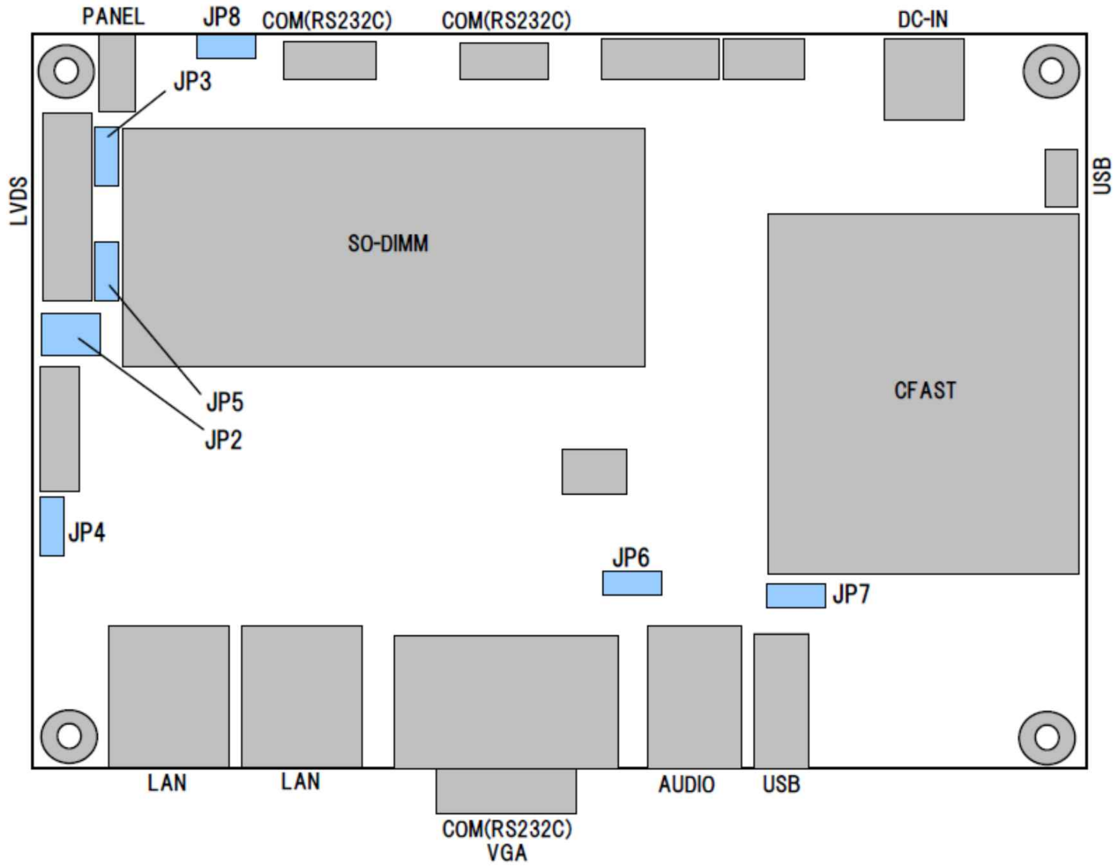
(5) LVDS1: LCD interface

Connector No.	Pin No.	Signal name	Remarks	Pin No.	Signal name	Remarks
LVDS1 LCD Interface	1	LVDS_VCC		2	LVDS_VCC	
	3	LVDS_VCC		4	LVDS_VCC	
	5	GND		6	GND	
	7	GND		8	GND	
	9	SELLVDS		10	N.C	
	11	N.C		12	N.C	
	13	GND		14	GND	
	15	GND		16	LVDS0_CLK+	
	17	LVDS0_CLK-		18	GND	
	19	LVDS0_D2+		20	LVDS0_D2-	
	21	GND		22	LVDS0_D1+	
	23	LVDS0_D1-		24	GND	
	25	LVDS0_D0+		26	LVDS0_D0-	
	27	GND		28	LVDS0_D3+	
	29	LVDS0_D3-		30	GND	

Junper Setting

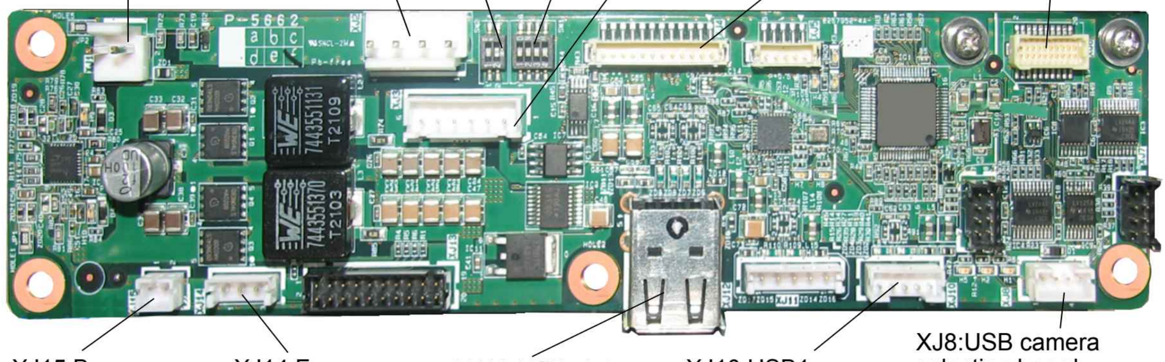
JP No.	Function	Selection	Position of jumper	Remarks
JP2	LVDS_VCC	3.3 V	* 1-3 2-4	
		5.0 V	3-5 4-6	
JP3	SELLVDS	High	1-2	Setting for pin 9 of LVDS connector The voltage level of "High" is LVDS_VCC.
		Low	2-3	
JP4	ENABKL	3.3 V	* 1-2	Pin 1 of BACKLIGHT connector. Voltage level setting of High.
		5.0 V	2-3	
JP5	LVDS0_D3+	1.8 V	* 1-2	Setting for pin 29 of LVDS connector.
		3.3 V	2-3	
JP6	Clear CMOS	Normal	* 1-2	
		Clear CMOS	2-3	
JP7	C fast PS voltage	3.3 V	* 1-2	
		5.0 V	2-3	
JP8	Power ON	Hardware Auto Power ON	* 1-2	Starts regardless the setting of Restore AC Power Loss of BIOS after power ON.
		N.C	2-3	

*: Default position



4.7.4 RCU I/F Board (P-5662*)

Appearance, Connector and DIP-SW locations



Function of board

1. TP-I/F board (P-5561*) control
2. Power supply and USB communication with CPU board
3. COmmunication with USB slots
4. Buzzer control
5. Printer control
6. Fan control

DIP-SW Setting

(1) SW1

SW1	Function	Selection	ON/OFF	Remarks
1	Not use		* OFF	
			ON	
2	Print mode	Inverted mode	* OFF	Handstand mode at CCW
		Handstand mode	ON	
3	Delay start	0 sec	* OFF	Delay start for DACS-G
		45 sec	ON	
4	Camera selection mode	Normal	* OFF	
		Reset at switch	ON	

*: Default setting

(2) SW2

SW2	Function	Select	ON/OFF	Remarks
1	Touch panel	4 wires	* OFF	
		5 wires	ON	
2	Serial output port	XJ7(DF11-8P)	OFF	
		XJ6(SHD-20P)	* ON	

*: Default setting

Connector classification

Connector No.		Function
XJ1	(2P)	Power supply pack (to main PS unit)
XJ2	(4P)	CPU board DC power(to JPWR1 on CPU board)
XJ3	(6P)	TP I/F board DC power
XJ4	(12P)	TP I/F
XJ6	(20P)	Printer I/F
XJ8	(4P)	USB camera selection signal
XJ10	(5P)	USB interface (to USB1 on CPU board)
XJ14	(3P)	Fan I/F
XJ15	(2P)	Buzzer I/F

Connector function in details

(1) XJ1: Power supply pack (to main PS unit)

Connector No.	Pin No.	Signal name	Remarks
XJ1 DC Power IN	1	DC+24V	
	2	GND	

(2) XJ2: CPU board DC power(to JPWR1 on CPU board)

Connector No.	Pin No.	Signal name	Remarks
XJ2 DC power OUT (to JPWR1 on CPU)	1	DC+5V	
	2	GND	
	3	DC+12V	
	4	GND	

(3) XJ3:TP I/F board DC power

Connector No.	Pin No.	Signal name	Remarks
XJ3 DC Power Output (For TP-I/F board)	1	+3.3V	
	2	GND	
	3	+5V	
	4	GND	
	5	+12V	
	6	GND	

(4) XJ4: TP-I/F

Connector No.	Pin No.	Signal name	Remarks
XJ4 TP-IF bord interface	1	+3.3V	
	2	GND	
	3	INT	
	4	CLK	
	5	MISO	
	6	MOSI	
	7	SCL	
	8	SDA	
	9	GND	
	10	VRMT	
	11	VBR	
	12	GND	

(5) XJ6: Printer I/F

Connector No.	Pin No.	Signal name	Remarks	Pin No.	Signal name	Remarks
XJ6 Printer I/F	1	N.C		2	N.C	
	3	N.C		4	N.C	
	5	N.C		6	N.C	
	7	RxD		8	TxD	
	9	GND		10	GND	
	11	GND		12	+5V	
	13	+5V		14	+5V	
	15	+5V		16	+5V	
	17	GND		18	GND	
	19	GND		20	+5V	

(6) XJ8: USB camera selection signal

Connector No.	Pin No.	Signal name	Remarks
XJ8 USB camera selection signal	1	+5V	
	2	OUT	
	3	GND	
	4	N.C	

(7) XJ10: USB interface (to USB1 on CPU board)

Connector No.	Pin No.	Signal name	Remarks
XJ10 USB Interface (to USB1 on CPU)	1	VBUS	
	2	GND	
	3	D-	
	4	D+	
	5	FG	

(8) XJ14: Fan I/F

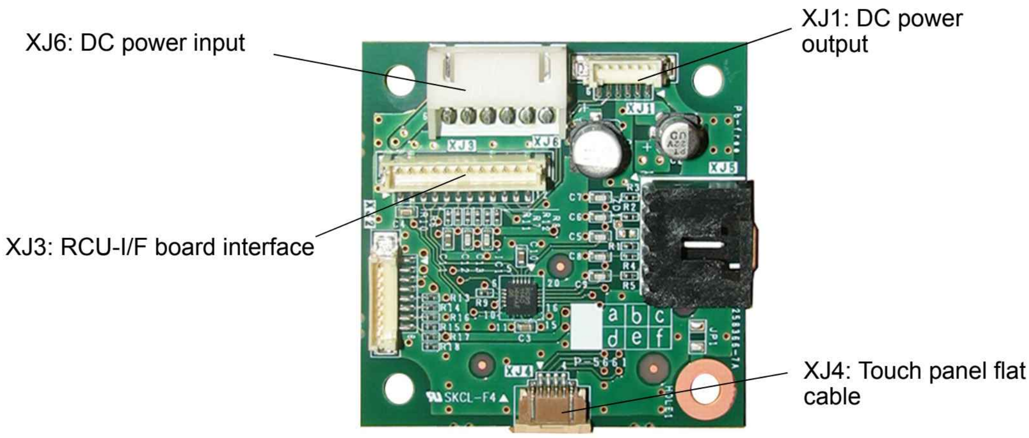
Connector No.	Pin No.	Signal name	Remarks
XJ14 Fan interface	1	+12V	
	2	IN(ALARM)	
	3	GND	

(9) XJ15: Buzzer I/F

Connector No.	Pin No.	Signal name	Remarks
XJ15 Bu7zzer interface	1	+5V	
	2	OUT	

4.7.5 TP I/F board (P-5661*)

Appearance of TP-I/F board (P-5561*)

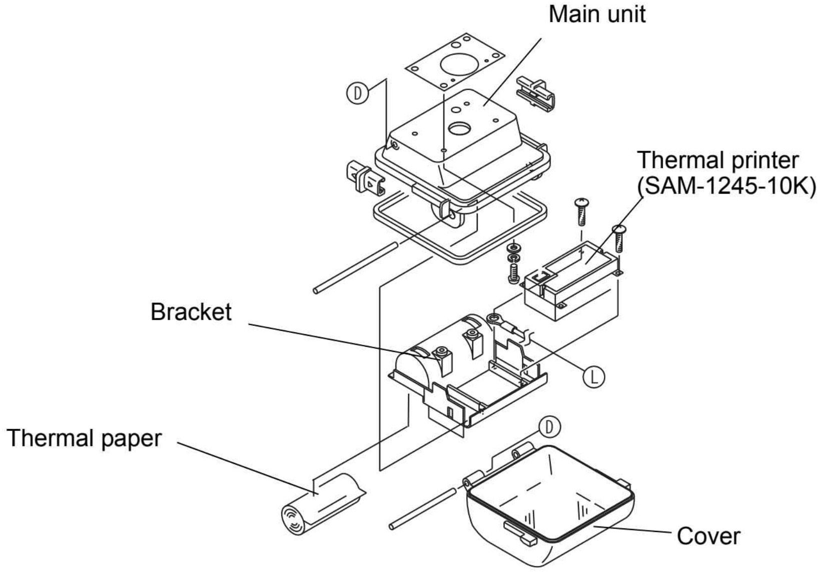


Functional description

1. Controls the signal sent from the touch panel and transmits it to CPU board via RCU/IF board.
2. Power supply to LCD unit

4.7.6 Printer Unit

Appearance of printer unit

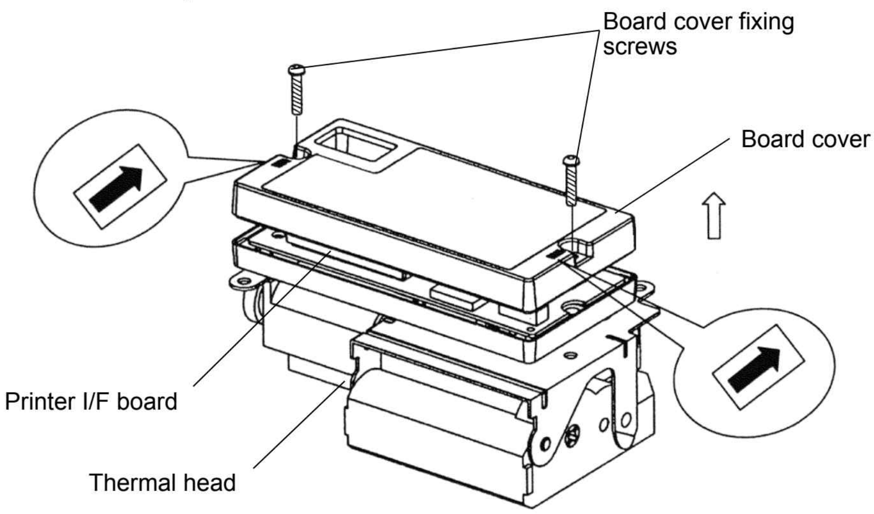


Function of unit

1. Printing statistical result totals
2. Printing presets and set values

4.7.6.1 Thermal Printer (SAM-1245-10K)

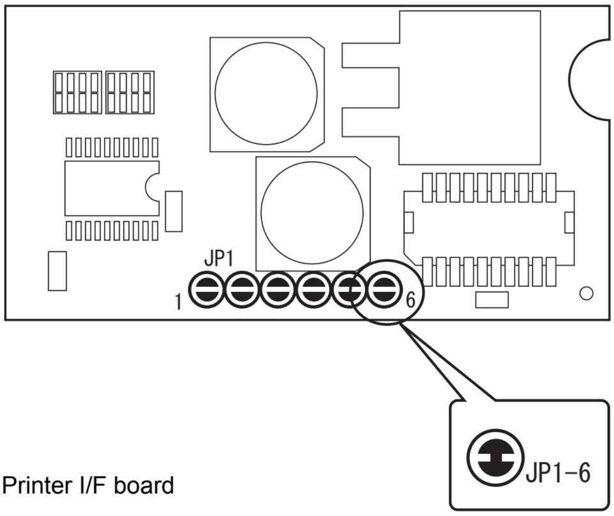
Appearance of thermal printer



Replacement of thermal printer

1. Open the door of printer main body.
2. Remove the roll paper for printing.

3. Remove two screws from the plastic bracket fixing the thermal printer and remove the bracket.
4. Reverse the bracket and remove the connector connected to the thermal printer.
5. Remove three screws fixing the thermal printer to the bracket, and remove the thermal printer.
6. Prepare a new thermal printer.
7. Remove two screws fixing the board cover, and remove the board cover.
8. Allow the function selection of the thermal printer.
9. Short-circuit the soldering jumpers for JP1-6 on the board as shown below.



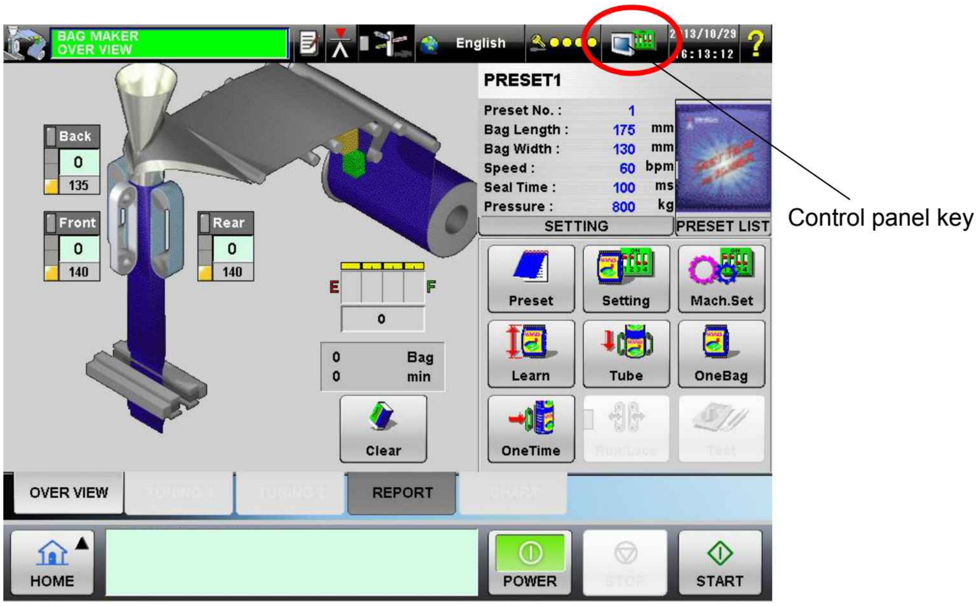
Printer I/F board

10. Attach the thermal printer in the reverse steps of the above 1 to 7.

NOTE

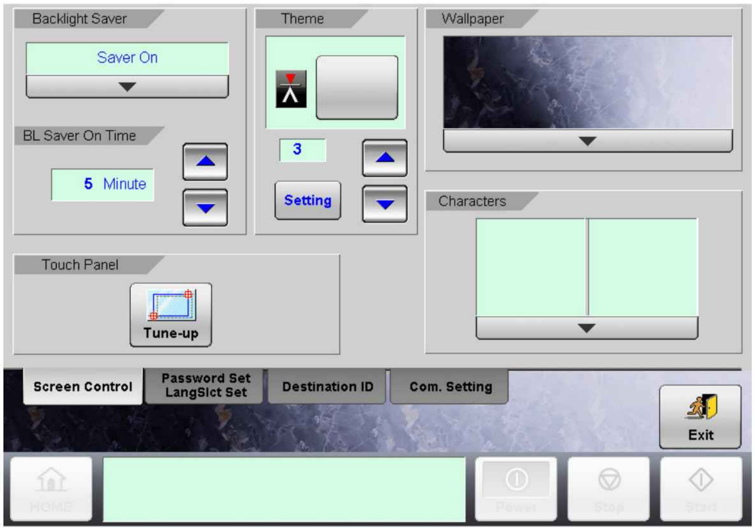
- As the soldering jumpers JP1-1 to JP1-6 of the new thermal printer are open, be sure to short ircuit the soldering jumper for JP1-6. Using the thermal printer without performing soldering jumper results in malfunction.
- The same type of thermal printer is used for the DACS-W-N model with the different location of oldering jumper. Refer to the Service Manual of the DACS-W-N model.

4.8 Control Panel Items (Maintenance Service)



[Main menu screen]

Touch .



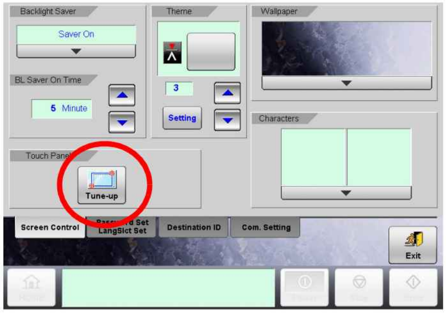
[Main Menu Screen with Control Panel Opened]

4.8.1 Display Control Menu Screens

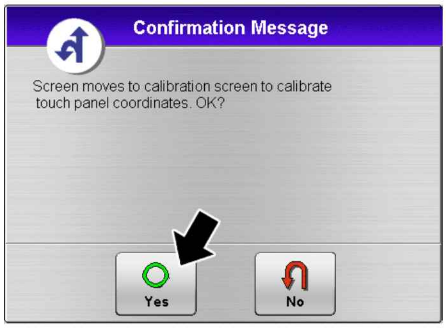
4.8.1.1 Touch Panel Coordinate Adjustment

TIP

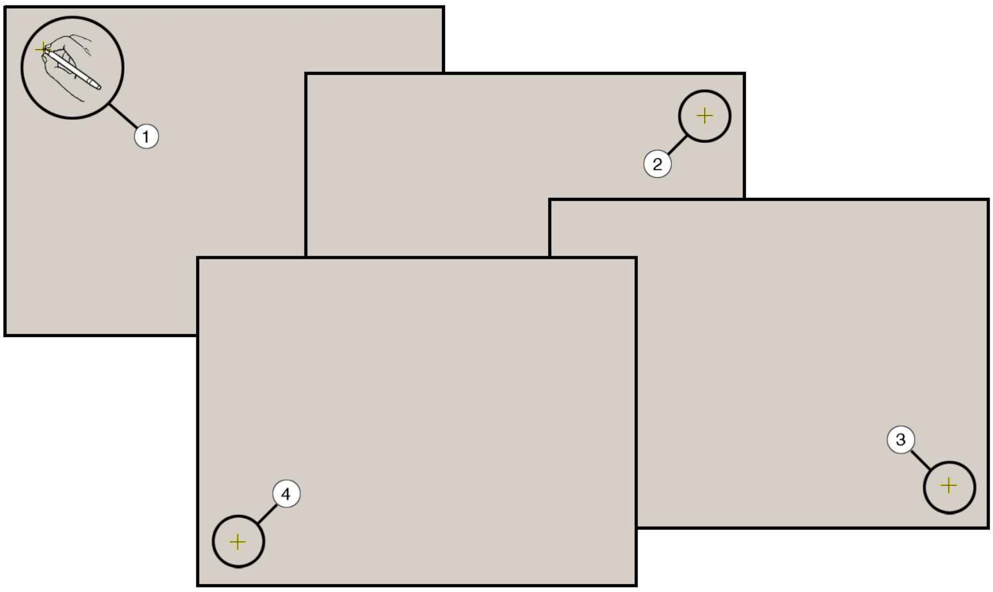
- When the other page such as "password setting" is opened right before the operation, touch the [Screen Control] menu.



1. Touch the [Tune-up] key on the [Display Control menu] screen.
2. The [Confirmation Message] appears, and touch the [Yes] key. The [Touchkit] screen appears.
3. Touch the [Cal 4 Point] key.



4. The following screens appear, and touch on the mark at each coordinate position with a ball point pen, etc.



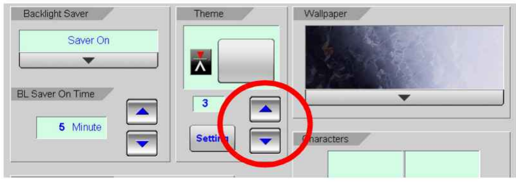
5. Touch on the coordinate position mark displayed automatically in order. After 4 points are touched, the [Confirmation Message] appears automatically.

4.8.1.2 Switching Theme Display (New Function)

Design of each key can be switched according to the theme: [1: Model R, 2: European model, 3: Model RV].

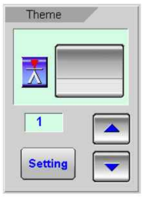
(This function is newly added to the current model.)

1. Touch the / key of [Theme].



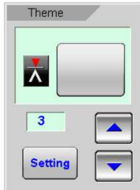
2. Switch the theme by selecting from [1: Model R, 2: European model, 3: Model RV].
3. Touch the [Setting] key.

   The display automatically restarts, with the theme switched.



1: Model R

2: European model



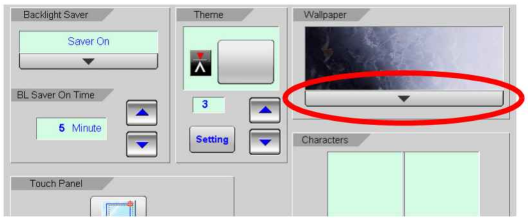
3: Model RV

4.8.1.3 Desktop Wallpaper Display (Additional Types Available)

The desktop wallpaper can be switched by selecting from several types. (Additional types are available from the current model).

The step is as follows:

1. Touch the pop-up key of [Wallpaper].



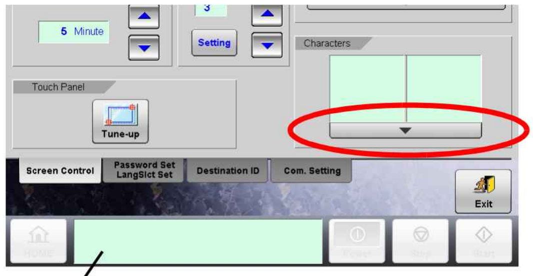
2. Select the wallpaper by touching it on the screen. The wallpaper is automatically switched to the selected option.

Touching the [Return] key enables to check that the wallpaper for the main menu display has been switched.

4.8.1.4 Character Display (Additional Types Available)

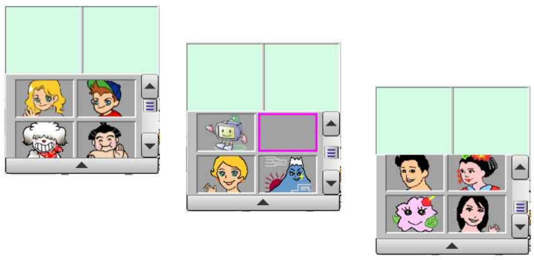
By setting the character display, condition of the machine is easy to grasp by the face look of the character.

1. Touch the pop-up key of [Character].
2. Select the character by touching it on the screen. The selected character is displayed on the [Status Display].

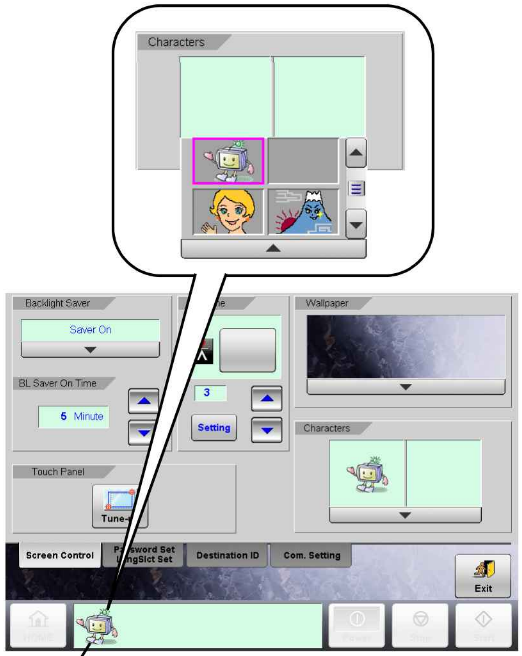


Status display

▶ The face look of the character shows the machine condition.



▶ When the machine condition is good, the character “smiles”, whereas when the machine condition has any problem, the character “cries”.



Status display

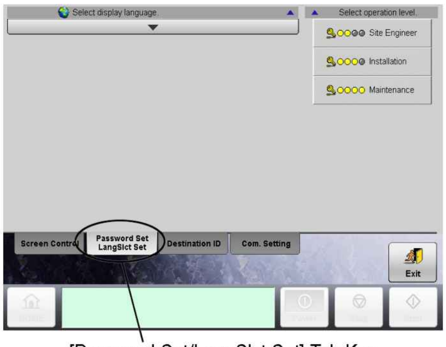
4.8.2 Password Set/Language Select Set Menu Screen

4.8.2.1 Language Select Setting

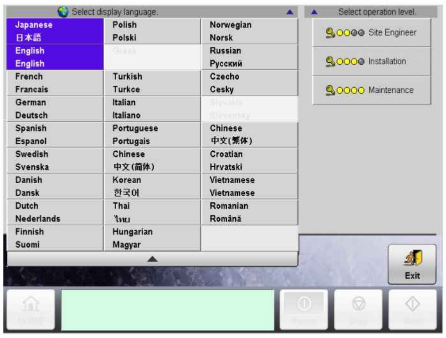
1. Touch the [Password Set/LangSlct Set] tab key. The Password Set/Language Select Set menu screen appears. Touch the [Select display language] pop-up key.
2. Select a display language.

TIP

- This model supports multiple languages as CCW-R (seven languages can be selected at the same time).



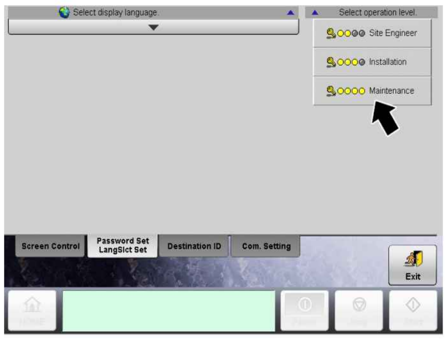
[Password Set/LangSlct Set] Tab Key

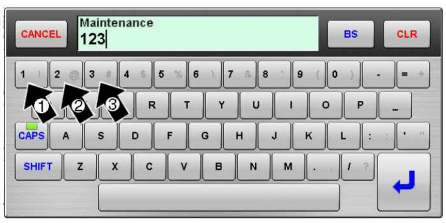


4.8.2.2 Password Setting

■ Touch the key to change the password on the [Select operation level].

[Example]



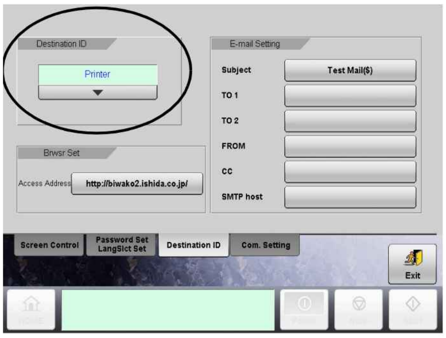


After entering password for change, touch the [Return] key.

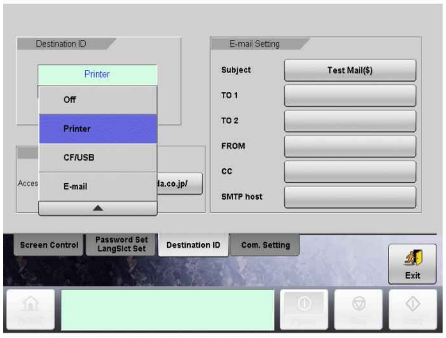
4.8.3 Destination ID Menu Screen

4.8.3.1 Destination ID

1. Touch the [Destination ID] tab key. The Destination ID menu screen appears.



2. Touch the [Destination ID] pop-up key, and select one from [Off, Printer, CF/USB or E-mail].



TIP

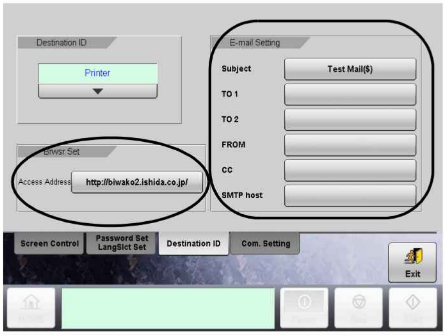
- Usually, select the [Printer].

4.8.3.2 Browser Setting

■ Set the [Access Address] accordingly.

4.8.3.3 E-mail Setting

■ Set each item of [Subject, TO 1...etc.] accordingly.



TIP

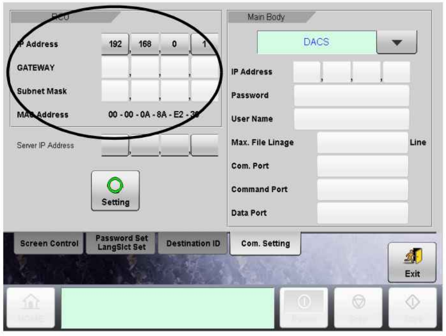
- This equipment should be connected to the network before setting.

4.8.4 Communication Set Menu Screen

■ Touch on the [Com. Setting] tab key. The [Communication Setting] menu screen appears.

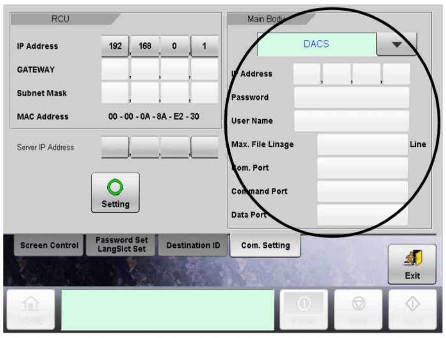
4.8.4.1 [RCU] Communication Setting

■ Set the [IP Address, GATEWAY, Subnet Mask] in RCU.



4.8.4.2 [Main Body] Communication Setting

■ Set the [IP Address, Password, User Name, etc.] in COM selected in the Main Body.

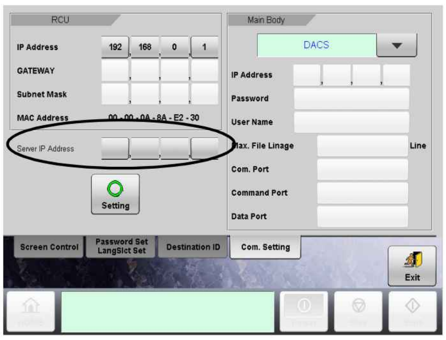


4.8.4.3 [Server IP Address] Setting

■ Set the Server IP Address.

■ When the machine is operated with i-STATION LINK2, input the same address as the RCU IP address and touch the [Setting] key.

The screen switches to the one with blue background. Be sure to wait for 30 seconds, and turn off the main power.



NOTE

- Be sure not to make an incorrect change. Otherwise, it may cause the communication failure.
- When the machine is operated with i-STATION LINK2, input the same address as the RCU IP address and touch the [Setting] key.
- When the screen switches to the one with blue background, be sure to wait for 30 seconds to turn off the main power.
- Turning off the main power at early timing may disable to switch to the correct IP address, resulting in “communication error”.

TIP

- At factory shipping, standard values are set; use them as they are.
- Change these values when interlocking with the computer after receiving the equipment.

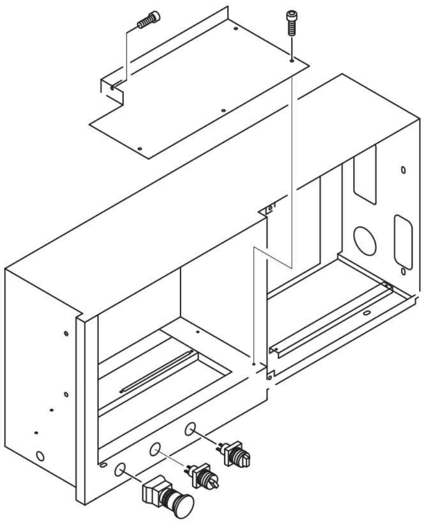
4.9 Electrical System

WARNING

- **Before replacing electrical components, make sure that the main power supply and motor power supply are turned OFF. Failure to do so can result in electrical shock.**

4.9.2 Relay Selector Switch Replacement

4.9.2.1 Front Switch Panel



1. Turn the main power OFF.
2. Swing the remote control unit.
3. Remove 5 screws and remove the switch plate.
4. Disconnect the cables that are connected to the switch.

NOTE

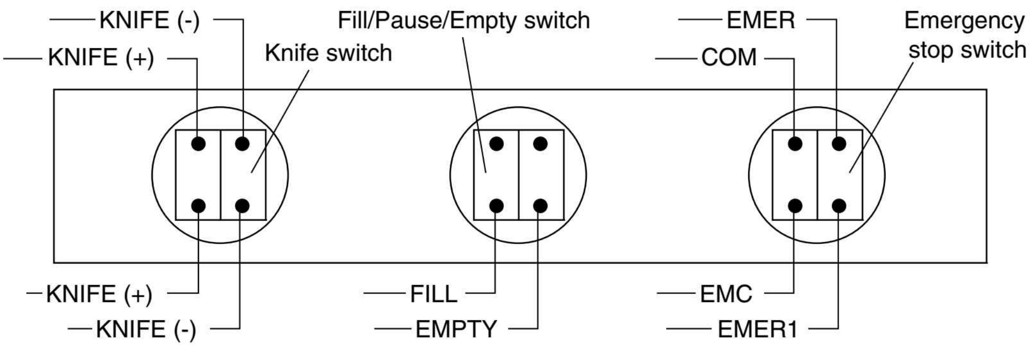
- Remove the cables in order, starting with the cable having the lowest number. (Refer to the tags attached to the cables.)
- Be sure wires are numbered.

5. Remove faulty switch.
6. Install the new switch.
7. Wire each switch.

NOTE

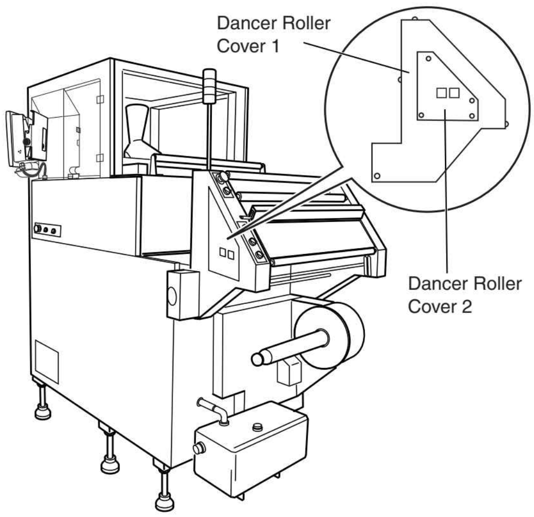
- The tags on the switch connecting cables are numbered as follows:

8. Reinstall all parts in the reverse order of removal.



4.9.2.2 Rear Switch Panel

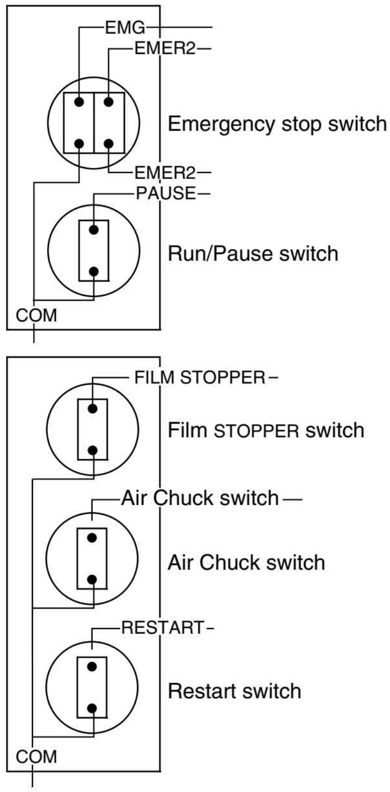
1. Turn the main power OFF.
2. Loosen the four bolts, and remove the rear switch panel from the guide arm.
3. Disconnect the cables that are connected to the faulty switch.



NOTE

- Remove the cables in order, starting with the cable having the lowest number. (Refer to the tags attached to the cables.)
- Be sure wires are numbered.

4. Remove faulty switch.
5. Install the new switch.



NOTE

- Refer to the illustration about the tag numbers on the switch connecting cables.

6. Reinstall all parts in the reverse order of removal.

4.9.3 MCU Board

1. Turn the main power OFF.
2. Open the front cover.
3. Pull the handle forward, and swing the back seal unit out.
4. Turn the former release lever to the rear, and remove the former.

CAUTION

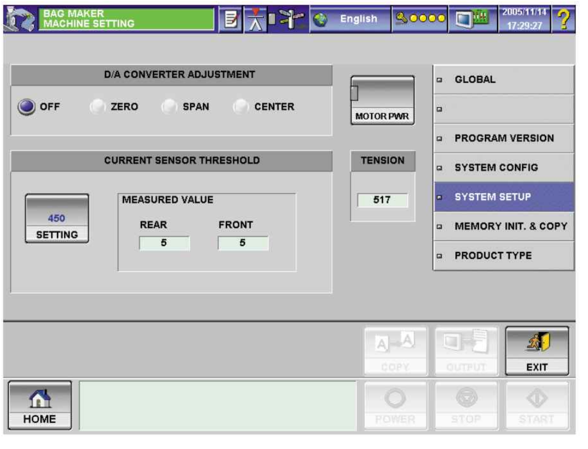
- **Be careful when handling the former. It is heavy.**

5. Remove the film roll.

NOTE

- Former and film must be removed to perform procedure!

6. Turn the main power supply ON.
7. Turn the motor power ON from Operation standby menu.
8. Enter the following password from the password screen:
   Password: 3720
   • The MACHINE SETTING screen will appear.

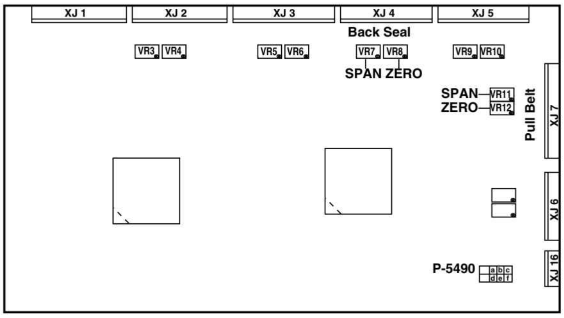


9. Press the following keys on the MACHINE SETTING screen to adjust D/A converter.
   • The left and right pull belt motor, back seal motor.
10. Press the ZERO key (ZERO adjust).
    Please check each servo driver display shows "3000". If a display shows out of "3000±2" range, please tune ZERO adjust trimmer for the servo driver on MCU Board with a small driver (Plastic point is preferable.) The display should show within "3000±2".

NOTE

- If a display does not show "3000±2" after tuning ZERO adjust trimmer, please refer to Appendix 2 and Appendix 3 and adjust the standard data. "500" in servo driver parameter Pr3.02. Then adjust ZERO / SPAN again.

MCU Board



11. Press the CENTER key.
12. Press the SPAN key (SPAN adjust).
    Please check each servo driver display shows "-3000". If a display shows out of "-3000±2" range, please turn SPAN adjust trimmer for the servo driver on MCU Board with a small driver. The display should show within "-3000±2".
13. If SPAN adjustment is performed.
    Please press the CENTER key again. Then press the ZERO key (ZERO adjust) and perform ZERO adjustment again as described in the above 11.
14. Press the STOP key after finishing each servo driver's ZERO (3000) and SPAN (-3000) adjustments.

NOTE

- If all values are within the standard range, press the EXIT key.

4.9.4 Replacing the Sensors

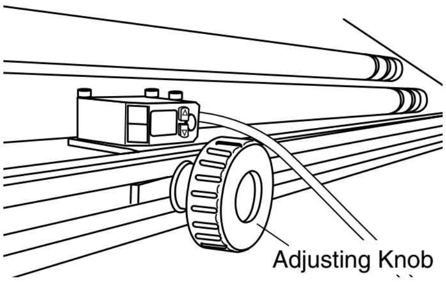
4.9.4.1 Eyemark adjustment

WARNING

- **Only authorized person can do this adjustment work.**

1. Turn the main power supply OFF.
2. The angle of the eye sensor (located at the rear of the machine) is measured when facing, and measuring upward from, the center of the sensor bar. Set the angle to the following value:

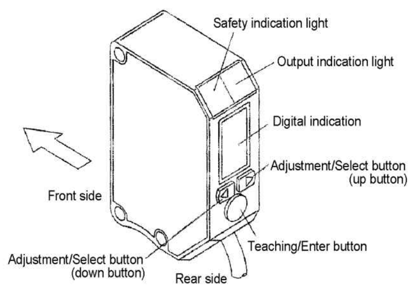
   Sensor angle: 0°



NOTE

- Sensor sensitivity may need to be adjusted by slightly changing the angle of the sensor bar when the eyemark sensor amplifier, which will be mentioned later, is adjusted.

Name of Each Part



3. Turn the main power supply ON.

DANGER

- **Wear protective clothing and protective gloves to touch electrical components. Failure to do so can result in electrical shock.**

< Adjusting eyemark sensor >

⚠ WARNING

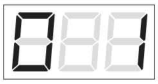
- **Exercise extreme caution when handling the eyemark sensor.**

1. Press the ▲▼ button for 2 seconds or longer in the RUN mode.

2. When the display is changed, press the ▲▼ button until the mark/color switch (CI) is displayed.

3. Press the [Teaching/Enter button].

4. Press the ▲▼ button to select the mark mode (0_I).

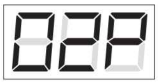


Two-point teaching in mark mode

1. Set the first point color (color for eyemark).

2. Press the [Teaching/Enter button].

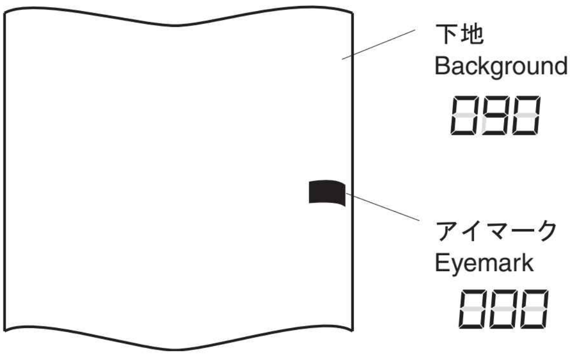
   ▶"02P" is displayed in the display.



3. Set the second point color (color for film ground).

4. Press the [Teaching/enter button].

   ▶When the teaching is performed normally, it returns to the RUN mode.
   When an error occurs, "0Er" is displayed.
   Press the [Teaching/Enter button] to perform the two-point teaching.



NOTE

- For the film, the ground color is "90" or more and the eyemark color is close to "0."